

AMATEUR TRANSCEIVER

MODEL : HR-2510

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PCB BOTTOM	(E23-7565)
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SERVICE INFORMATION MANUAL

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PARTS ASS'Y TOP	(E24-7574)
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Note: Circuit design and specifications are subject to change without notice.

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1	1. SPECIFICATIONS

1. SPECIFICATIONS

AMATEUR RADIO MODEL : HR2510 (UT-550B)

GENERAL

1. Channel and Frequency Range	: A Band 28.0000 ~ 28.4999 MHz B Band 28.5000 ~ 28.9999 MHz C Band 29.0000 ~ 29.4999 MHz D Band 29.5000 ~ 29.6999 MHz
2. Crystal	: 3
3. Microphone	: 500 Ohm, Dynamic Type with Channel up down
4. Speaker	: 8 Ohm, 3 W
5. Antenna Connector	: M Type
6. Jacks & Connectors	: MIC 5P, DC Power 2P, 9 PIN ACC CONNECTOR
7. Controls	: Mode Selector CW-LSB-USB-AM-FM, MIC Gain, TX, DIM, SCAN, SPAN, Meter Select, Beep, Band, F-Lock, CH up/down, PA, NB, Frequency Knob, ON/OFF Power Switch & Volume, Auto SQ & Squelch, RF Gain, RIT, SWR/CAL
8. Illumination	: Band No. Indication LCD Channel No. Indication LCD Frequency Indication LCD S/R, MOD, SWR Indication LCD
9. LCD Meter	: RF Power Indication, Signal Strength, Modulation, CAL, SWR
10. Size	: 186 mm (W) x 263 mm (D) x 62 mm (H)
11. Weight	: 1.9 kg
12. Accessories	: DC Power Cable with Fuse Microphone & Mic Hanger Mounting Bracket with Screws Short Plug 9 pin Accessory Plug
13. Display Box Size	: 376 (W) x 230 (D) x 110 (H)
14. Display Box Weight	: 2.75 kg (include Unit and ACC.)
15. Quantity / Shipping Carton	: 6 pcs.
16. Size / Shipping Carton	: 730 (W) x 416 (D) x 270 (H)
17. Weight / Shipping carton	: 18 kg (Per 6 pcs.)
18. Case Mark	: UCA Standard Style

MEASUREMENT CONDITIONS

1. Power Source	: 13.8 V (DC)
2. Antenna Impedance	: 50 Ohm
3. Test Temperature	: 25°C
4. AM Modulation Frequency	: 1 kHz
5. FM Modulation Frequency	: 1 kHz
6. SSB Modulation Frequency, Two Tone	: 500 Hz & 2400 Hz
7. Mean Signal Input Level	: 1000 μ V
8. Reference AM Modulation Percentage	: 1 kHz, 30%
9. Reference FM Deviation	: 1.5 kHz
10. Reference Audio Output Power	: 0.5 W
11. Audio Frequency	: 1 kHz
12. Audio Output Load	: 8 Ohm Resistive

TRANSMITTER SECTION

<u>ITEMS</u>		<u>UNIT</u>	<u>NOMINAL</u>	<u>LIMIT</u>
1. Frequency Tolerance at 25°C (5 minutes after Switch on)	AM	Hz	± 300	± 1500
	SSB	Hz	± 300	± 1500
	FM	Hz	± 300	± 1500
	CW	Hz	± 300	± 1500
2. Carrier Power	AM	W	10.0	8.0 ~ 14.0
	FM	W	10.0	8.0 ~ 14.0
	CW	W	21.0	18.0 ~ 26.0
3. PEP Power, Two Tone, SSB		W _{pep}	21.0	18.0 ~ 26.0
4. Spurious Harmonic Emission	AM	dB	-50	-40
	SSB	dB	-50	-40
	FM	dB	-50	-40
	CW	dB	-50	-40
5. Carrier Suppression	SSB	dB	-55	-40
6. Unwanted Sideband Suppression (at 2500 Hz 4 W _{pep} 16 dB up Single tone)				
	SSB	dB	-45	-35
7. Battery Drain at No Modulation	AM	mA	3000	4500
	SSB	mA	800	1500
	FM	mA	3000	4500
	CW	mA	5000	6500
8. Battery Drain				
	AM : MAX Modulation	mA	3000	4500
	SSB : MAX W _{pep} , Two Tone	mA	3000	4500
	FM : MAX Modulation	mA	3000	4500
9. Modulation Frequency Response (1 kHz, 0 dB Reference)				
	CW : TX Mode, No Key Down	mA	800	1200
Lower at 450 Hz				
	AM	dB	-4	-10
	SSB	dB	-4	-10
	FM	dB	-4	-10
	Upper at 2.5 kHz			
	AM	dB	-4	-10
	SSB	dB	-4	-10
10. Microphone Sensitivity				
	FM	dB	-4	-10
AM : For 50% Modulation		mV	1.0	2.0
	SSB : For 10 W _{pep} Output	mV	1.0	2.0
	FM : For 1 kHz Deviation	mV	1.5	3.0
11. AMC Range				
	AM : 50 ~ 100% Modulation	dB	50	40
SSB : 16 ~ 26 W _{pep}		dB	50	40
12. MIC Gain (ATTENATE)		dB	-10	-6

RECEIVER SECTION (NB Switch OFF)

1. Max. Sensitivity	AM	uV	0.5	2.0
	CW/SSB	uV	0.25	1.0
2. Sensitivity for 10 dB S/N	AM	uV	0.5	2.0
	CW/SSB	uV	0.25	1.0
FM : 20 dB S/N	FM	uV	0.5	3.0
3. AGC Figure of Merit, 50 mV for 10 dB Change in Audio Output	AM	dB	80	70
	CW/SSB	dB	80	70
4. Overall Audio Fidelity at 6 dB down				
Upper Frequency	AM	Hz	2000	1500 ~ 4000
	SSB	Hz	3000	2000 ~ 4500
	FM	Hz	1500	1000 ~ 3000
Lower Frequency	AM	Hz	300	100 ~ 500
	SSB	Hz	300	100 ~ 500
	FM	Hz	300	100 ~ 500
5. Adjacent Channel Selectivity (10 kHz, 1 GEN)	AM	dB	70	60
6. Maximum Audio Output Power	AM	W	4.0	2.5
	CW/SSB	W	4.0	2.5
	FM	W	4.0	2.5
7. Audio Output Power at 10 % THD	AM	W	2.5	2.0
	CW/SSB	W	2.5	2.0
	FM	W	2.5	2.0
8. RF Gain	AM	dB	55	30 ~ 70
	CW/SSB	dB	55	30 ~ 70
	FM	dB	55	30 ~ 70
9. S/N Ratio at 1 mV Input	AM	dB	30	25
	FM	dB	35	25
10. Squelch Sensitivity at Threshold	AM	uV	0.5	2.0
	CW/SSB	uV	0.5	2.0
	FM	uV	0.5	2.0
11. Squelch Sensitivity at Tight	AM	uV	1000	250 ~ 4000
	CW/SSB	uV	1000	250 ~ 4000
	FM	uV	1000	250 ~ 4000
12. S Meter Sensitivity at "S-9" (No Modulation)	AM	uV	100	25 ~ 400
	CW/SSB	uV	100	25 ~ 400
	FM	uV	100	25 ~ 400
13. Image Rejection Ratio	AM	dB	65	50
	CW/SSB	dB	65	50
	FM	dB	65	50
14. i-f Rejection Ratio	AM	dB	65	50
	CW/SSB	dB	65	50
	FM	dB	65	50

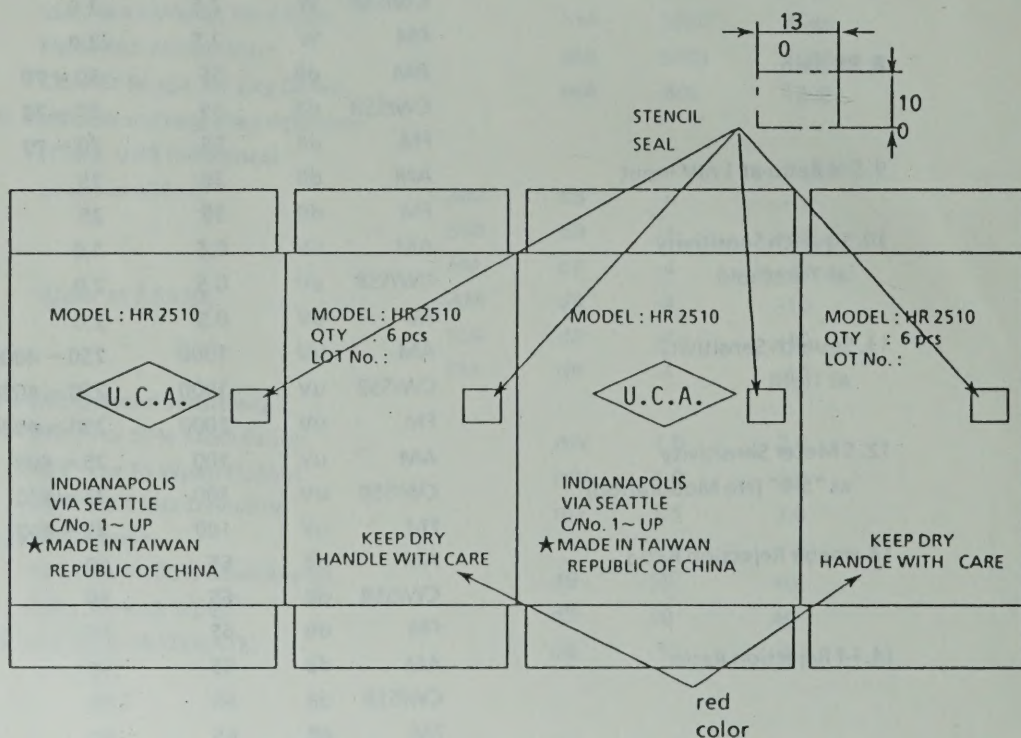
ITEMS		UNIT	NOMINAL	LIMIT
15. Oscillator Dropout Voltage	AM	V	9	11
	CW/SSB	V	9	11
	FM	V	9	11
16. Battery Drain at No Signal	AM	mA	500	800
	CW/SSB	mA	500	800
	FM	mA	500	800
17. Battery Drain at Max. Audio Output	AM	mA	1000	1500
	CW/SSB	mA	1000	1500
	FM	mA	1000	1500
18. Clarifier Range	AM	kHz	± 3.0	$\pm 2.2 \sim \pm 4.0$
	CW/SSB	kHz	± 3.0	$\pm 2.2 \sim \pm 4.0$
	FM	kHz	± 3.0	$\pm 2.2 \sim \pm 4.0$

PUBLIC ADDRESS

ITEMS	UNIT	NOMINAL	LIMIT
1. Output Power at 10% Distortion	W	2.5	2

NOTE :

- (1) Shall have reverse polarity protection and operable with negative ground only.
- (2) Operating Power Voltage : 13.8V + 15 - 20% DC
- (3) Standard Operating Temperature : $-10^{\circ}\text{C} \sim +50^{\circ}\text{C}$
- (4) Storage Temperature : $-30^{\circ}\text{C} \sim +60^{\circ}\text{C}$



2. ALIGNMENT PROCEDURES

ALIGNMENT OF TRANSMITTER PORTION

1. Test Equipment Required

DC Power Supply (13.8V) more than 10A	
AF S.S.G. AM FM (1kHz, 500Hz and 2400Hz)	
RF VTVM	Oscilloscope
DC AM METER	RF Power Meter
AF VTVM	Dummy Load (50 ohm)
FM Linear Detector	

2. Preparation for Alignment

VR112	:	Clockwise	PA SW	:	OFF
VR113	:	Counter Clockwise	METER SW	:	RF
VR103	:	Clockwise	Mic Gain SW	:	OFF
SWR/CAL	:	Middle Position	TX SW	:	OFF
FREQ.	:	28.000 MHz			

3. Alignment Procedure

Step	Preset to	Adjustment	Remarks
1	Mode : USB No.mod.	VR112	Remove the B002(PB-100) from Main PCB. Connect a DC AM Meter(+) to TP4, (-) to TP3. Adjust VR112 for 50 mA reading on the DC AM Meter.
2	Ditto	VR113	Connect the DC AM Meter(+) to TP4, (-) to TP2. Adjust VR113 for 50 mA reading on the DC AM Meter.
3	1mV modulation	L111	Disconnect the DC AM Meter. Reinstall the B002 to the Main PCB. Connect a RF Power meter to the ANT.jack.Connect a RF VTVM, an oscilloscope and an FM Linear Detector across an RF Dummy load to the RF Power meter. Adjust L111 for maximum reading on the RF VTVM. During this step, set the AF oscillator so that the output is less than 20V. Repeat this step two times.
4	30mV modulation	VR104	Adjust VR104 for 32.5V reading on the RF VTVM.
5	Ditto	VR106	Adjust VR106 so that the carrier leakage at USB and LSB become minimum and almost equal.
6	Mode : CW Non Mod.	VR103	Connect a SW to between Pin 8 and 9 of ACC connector. When turn on the SW, adjust VR103 for 21W reading on the power meter.

ALIGNMENT OF P. L. L. AND CARRIER OSCILLATOR PORTION

1. Test Equipment Required

DC Power Supply (13.8V)
DC Voltmeter
Dummy Load (50 ohm)

Oscilloscope
Frequency Counter

2. Preparation for alignment

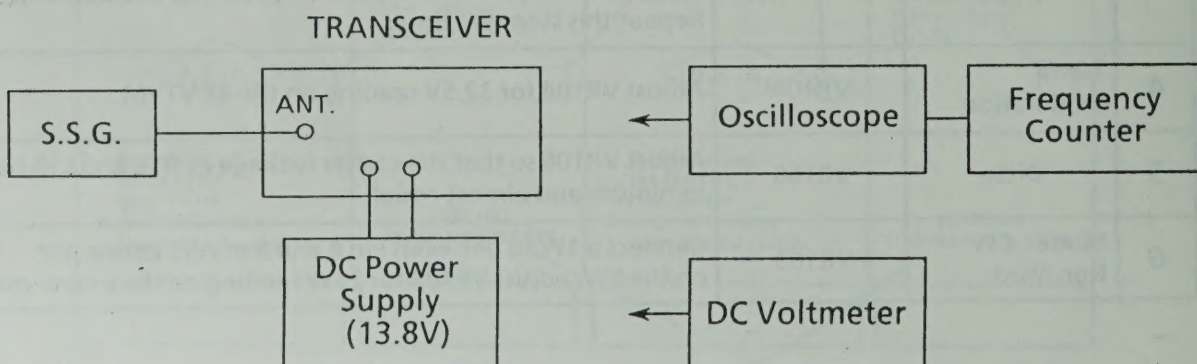
PA SW : OFF
RIT : Middle Position

MODE SW : AM
TX : OFF

3. Alignment Procedure

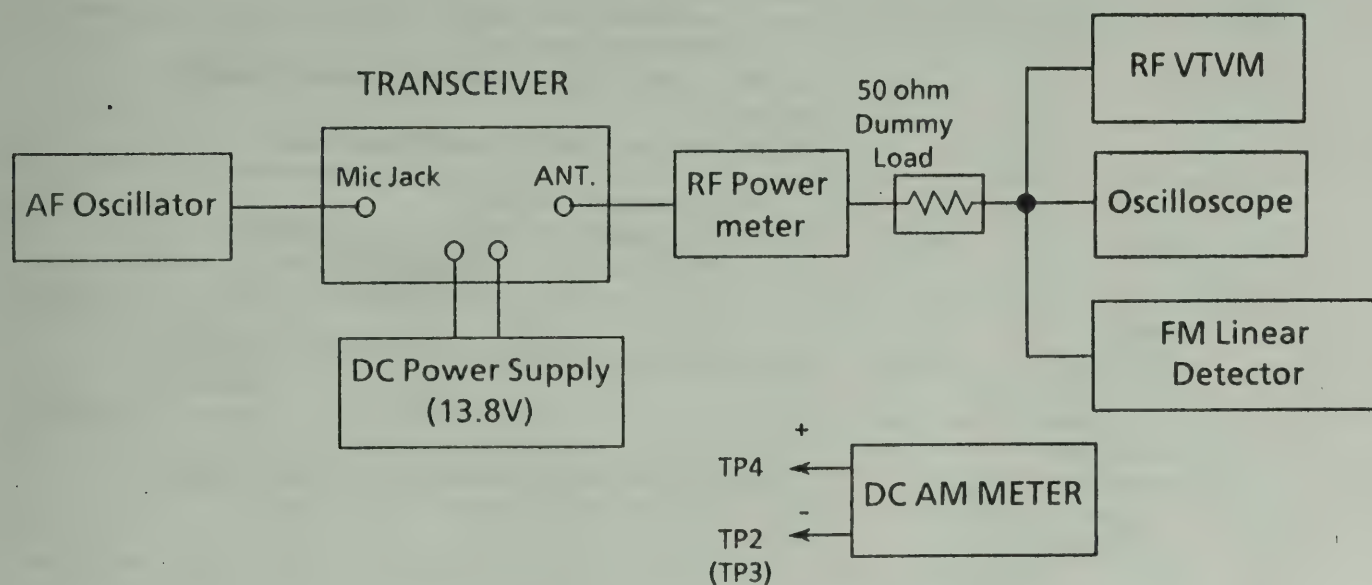
Step	Preset to	Adjustment	Remarks
1	Mode: RX,AM SG : 28MHz	L315	Connect an oscilloscope to TP306. Adjust L315 for 6.200MHz $\pm$ reading on the oscilloscope.
2	Ditto	L318	Connect the oscilloscope to TP304. Adjust L318 for 38.695MHz $\pm$ 20 Hz reading on the oscilloscope.
3	Mode: RX,AM SG:29.6999MHz	L317	Connect a DC voltmeter to TP303. Adjust L317 for 6.5 $\pm$ 0.1V reading on the DC Voltmeter.
4	Mode: RX,CW SG : 28MHz	L117	Connect the oscilloscope to TP1. Adjust L117 for 10.6950MHz $\pm$ 20 Hz reading on the oscilloscope.
5	Mode: RX,LSB	L118	Adjust L118 for 110.6925 MHz (-40Hz to + 0Hz) reading on the oscilloscope.
6	Mode:RX,USB	L116	Adjust L116 for 10.6975 MHz $\pm$ 20 Hz reading on the oscilloscope.
7	Mode:RX,USB	VR111	Connect the oscilloscope to TP5. Adjust VR111 for 38.6975MHz $\pm$ 20Hz reading on the oscilloscope.

TEST EQUIPMENT CONNECTION



Step	Preset to	Adjustment	Remarks
7	Mode : AM Non Mod	VR107	Adjust VR107 for 10W reading on the RF power meter.
8	Ditto	VR117	Adjust VR117 so that "9" LCD just lights on.
9	1kHz, 30mV Modulation	VR114	Adjust VR114 to obtain the 85% negative reading on the oscilloscope.
10	1kHz, 1mV Modulation INDIC : MOD	VR115	Adjust VR115 so that "9" LCD just lights on.
11	Mode : FM 1kHz, 30mV Modulation	VR105	Adjust VR105 for ± 3 kHz deviation reading on the FM Linear Detector.
12	Mode: CW No Mod. Vol. : Max.	VR116	Connect an AF VTVM across a dummy load (8ohm) to between Pin 1 and Pin 2 of ACC connector. When turn on the SW, adjust VR116 for 0.4V reading on the AF VTVM.

TEST EQUIPMENT CONNECTION



ALIGNMENT OF RECEIVER PORTION

1. Test Equipment Required

AF VTVM
Oscilloscope
S.S.G.(28.000MHz , 1kHz, 30% Mod. and 50-ohm Impedance)

DC Power Supply (13.8V)
Dummy Load (8 ohm)

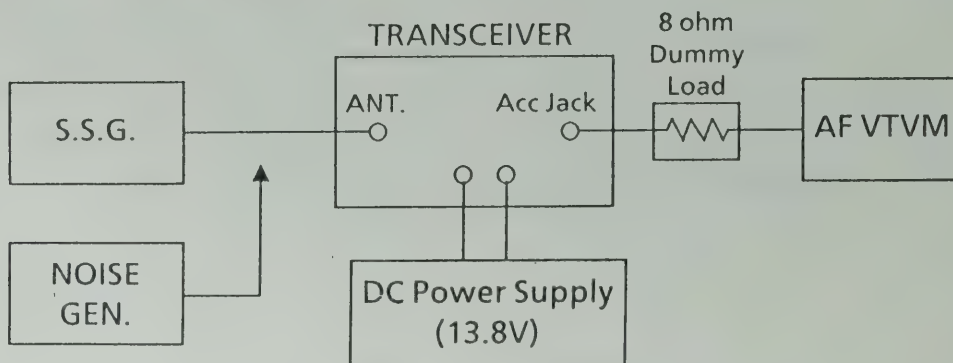
2. Alignment Procedure

NB SW	: OFF	Mode	: AM
PA SW	: OFF	Squelch	: Min. (Auto SQ OFF)
Beep SW	: OFF	RF Gain	: MAX
VOLUME	: MAX	TX	: OFF
RIT	: Middle position		

3. Alignment Procedure

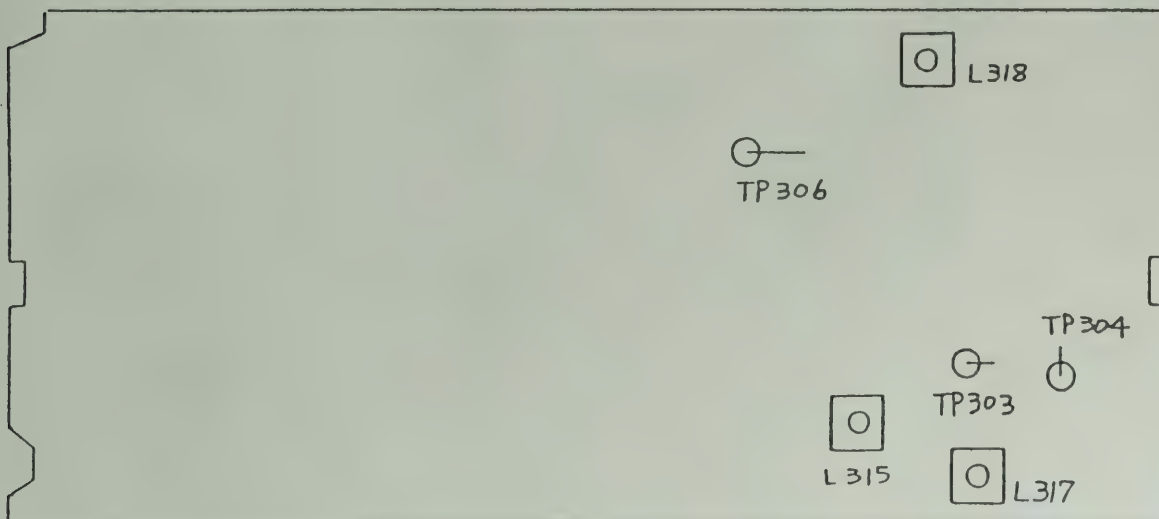
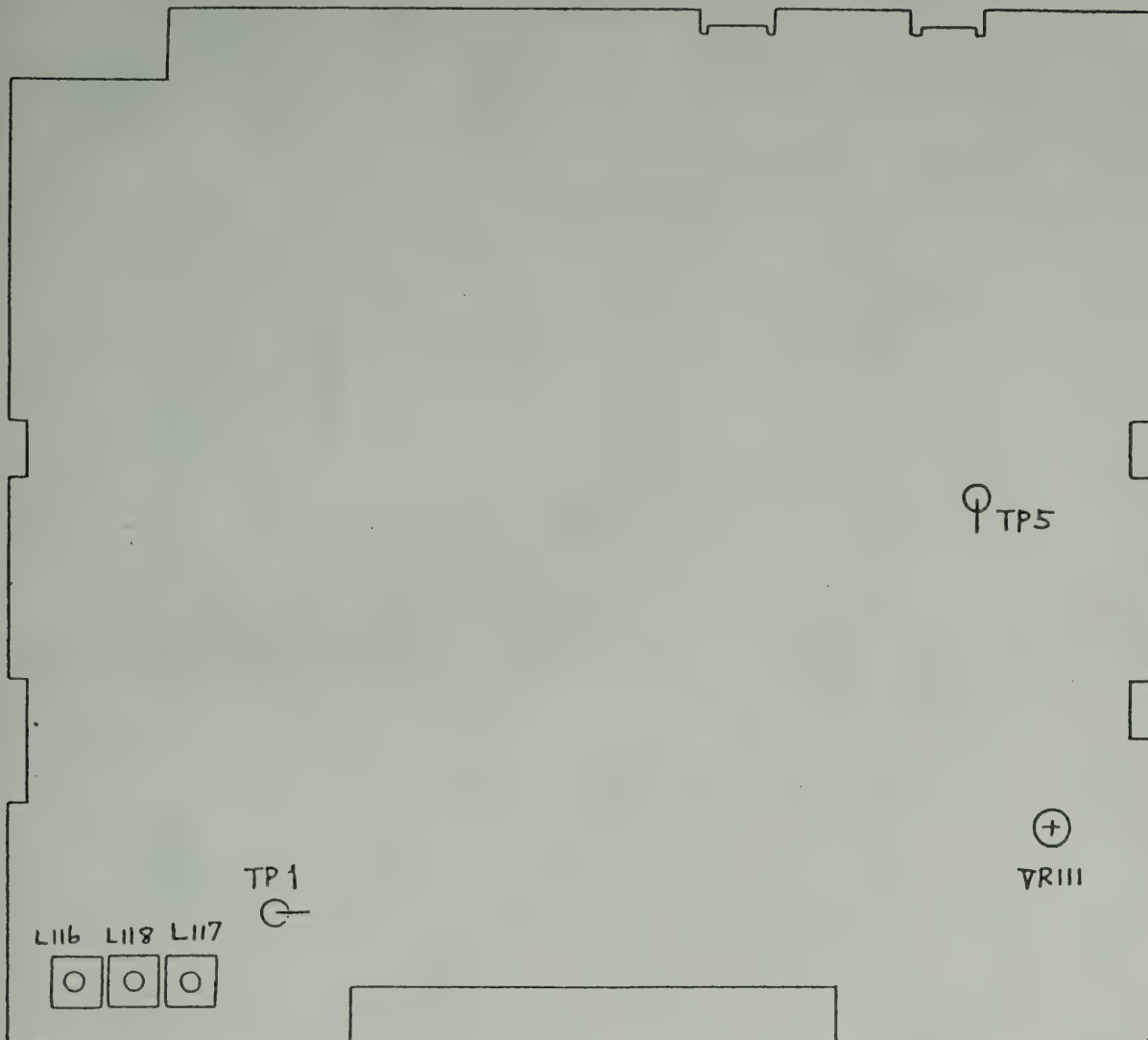
Step	Preset to	Adjustment	Remarks
1		L101, L104, L113, L115 and L105	Alignment of sensitivity Adjust coils for maximum reading on the AF VTVM.(During this step, set the S.S.G.attenuator so that the standard output is less than 0.5W (2V / 8ohm).
2		VR102	Alignment of Squelch Set the output of S.S.G. to 66 dB $\pm$ 2 dB and squelch to maximum. Adjust VR102 so that the squelch just breaks.
3		VR101	Alignment of S- Meter Set the output of S.S.G. to 46 dB, no modulation. Adjust VR101 so that "9" LCD just lights on.
4	Mode : FM S.S.G.: 1mV (1.5kHz Dev.)	L401	Adjust L401 for maximum reading on the AF VTVM.
5	Mode : AM Noise GEN. to the ANT.Jack	L203	Adjust L203 for minimum reading on the AF VTVM. (Noise GEN. OUTPUT : 50Hz SQ WAVE 2Vp-p)

TEST EQUIPMENT CONNECTION

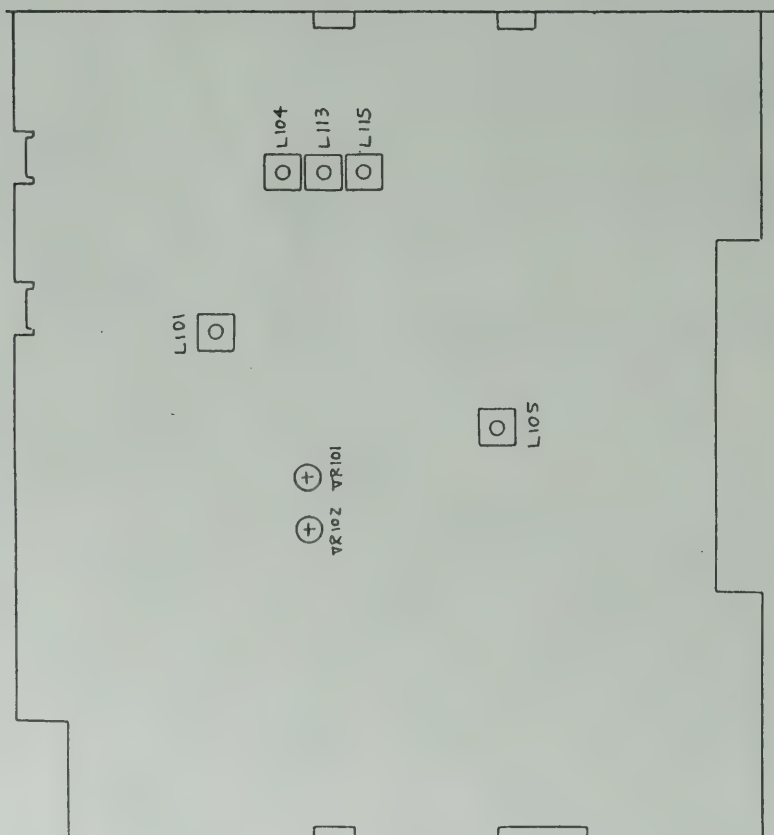


3. ALIGNMENT POINTS

Alignment for PLL and Carrier Oscillator Portion

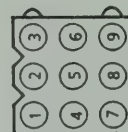


Alignment for Receiver Portion

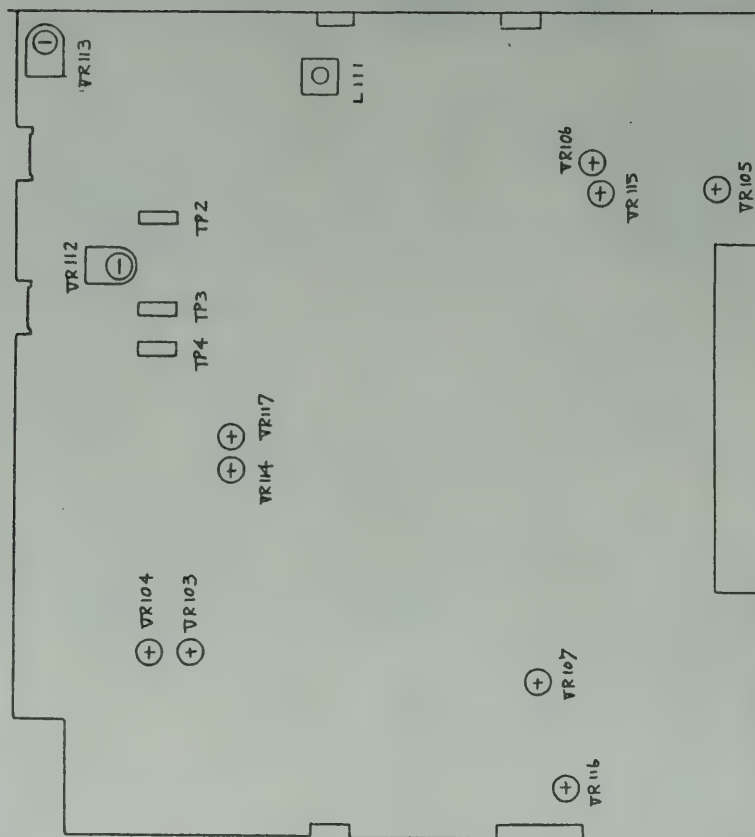


CONNECTION FOR Rx ALIGNMENT

ACCPUG

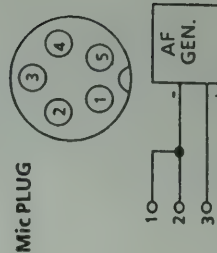
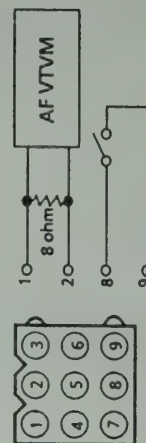


Alignment for Transmitter Portion



CONNECTION FOR Tx ALIGNMENT

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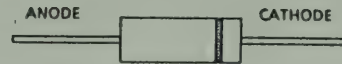


4. SEMICONDUCTOR LEAD IDENTIFICATION

DIODE



1N4148
HZ5C-1
HZ3B3
1N60 AM
1N60 P
1N4003



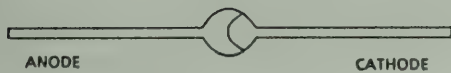
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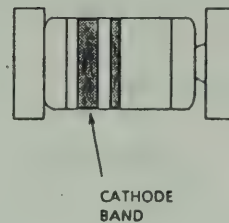
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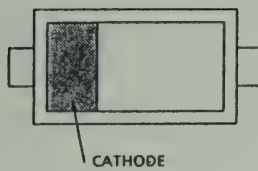
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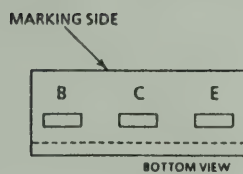
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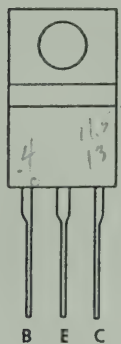
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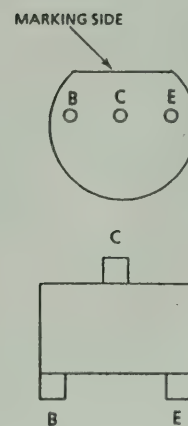
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25C2166-C



MRF-477
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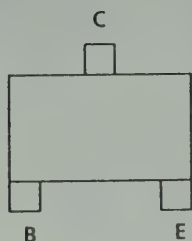


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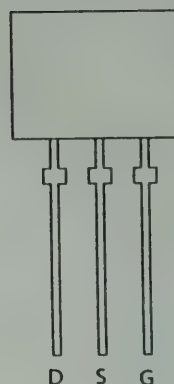
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TRANSISTOR

TRANSISTOR



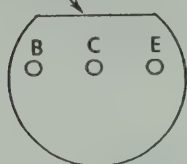
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2SC3121



G: GATE
S: SOURCE
D: DRAIN

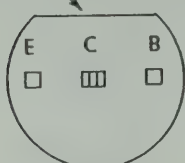
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MARKING SIDE



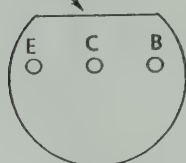
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MARKING SIDE

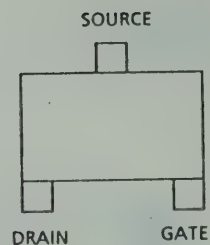


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MARKING SIDE



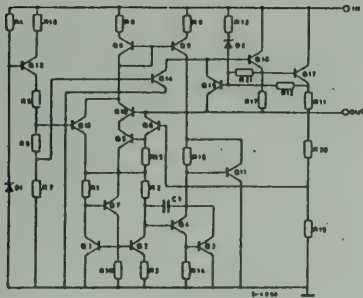
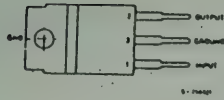
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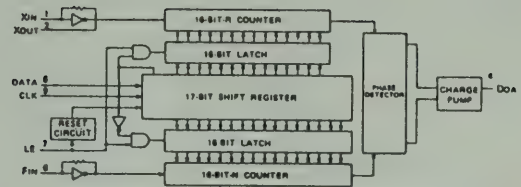
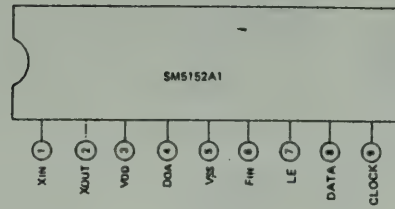
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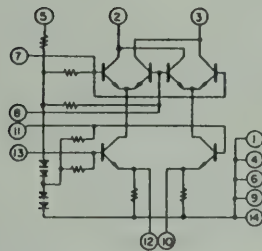
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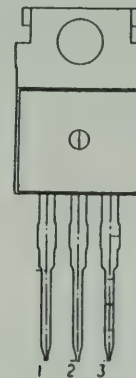
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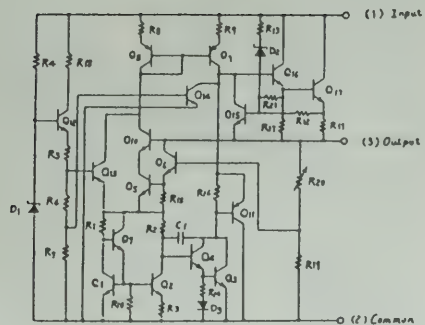
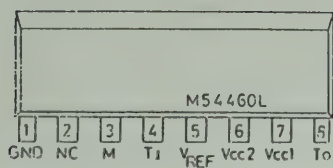
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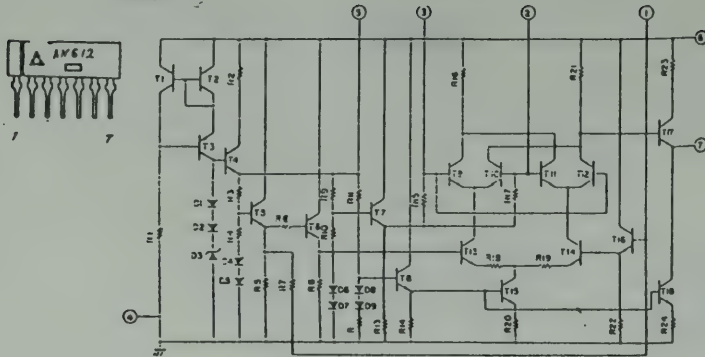
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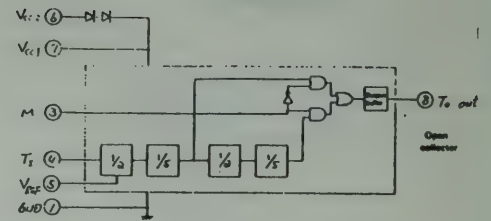
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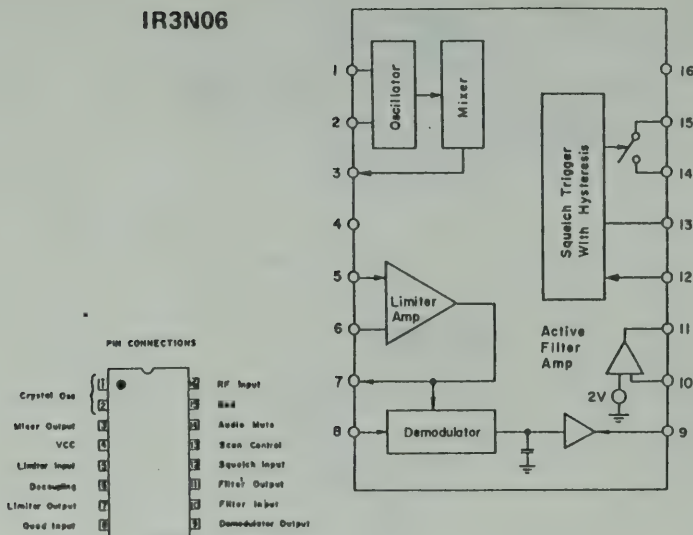
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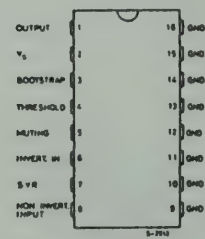
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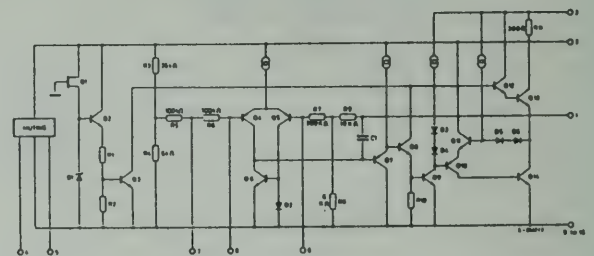
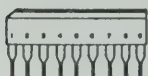
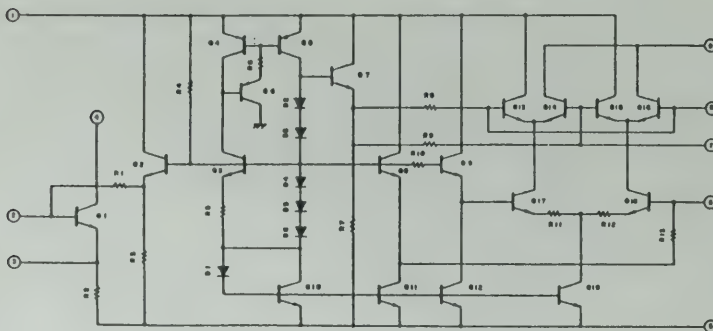


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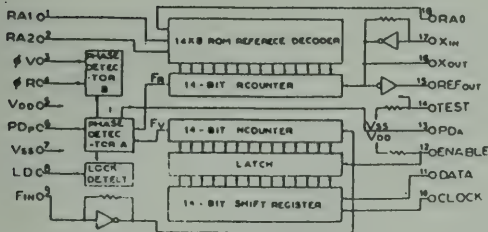
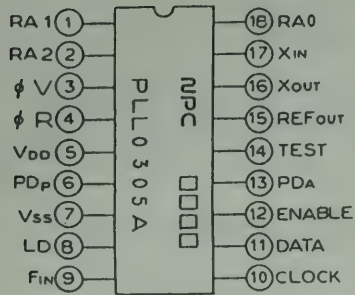


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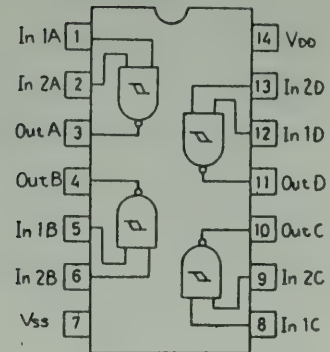
BALANCED MODULATOR-DEMODULATOR



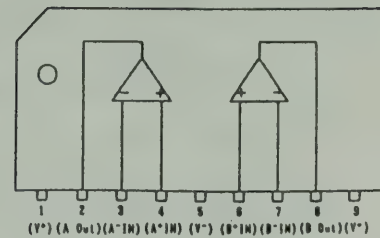
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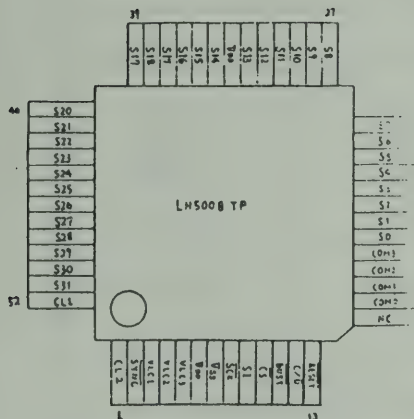
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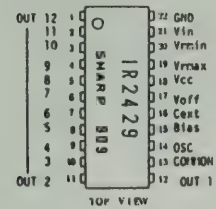
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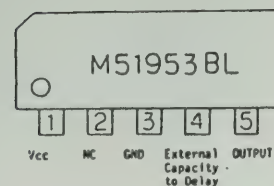
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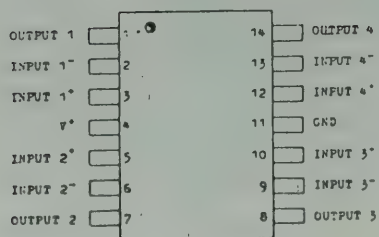
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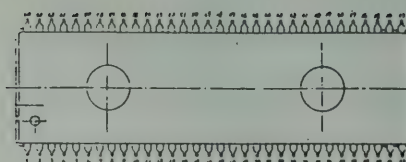


NJM2902N



(TOP VIEW)

UC1201



UC1201

Pin No.	Name	Pin No.	Name	Pin No.	Name	Pin No.	Name
1	LCD1 (SCK)	17		33	BEEP	49	
2	LCD2 (SI)	18	CWFiL	34	AMATEUR	50	
3	LCD3 (CS)	19		35	GND	51	
4	LCD4 (C/B)	20	BPF2	36	LÖCD	52	
5	LCD5 (RESET)	21		37	VCXÖ1	53	
6	RF	22	FKNOB1	38	VCXÖ2	54	ABSEL
7	MOD	23	FKNOB2	39	VCXÖ3	55	S&U
8	CAL	24	AM	40	VCXÖ4	56	STÖP
9	SWR	25	USB	41	VCXÖ5	57	BPF3
10	RÖW1	26	LSB	42	VCXÖ6	58	WATCHD
11	RÖW2	27	CW	43		59	TÖNE
12	RÖW3	28	DATA	44	PÖW OFF	60	BPF1
13		29	CLK	45	MUP	61	LCD6 (BUSY)
14	CÖL1	30	STB1	46	MDÖWN	62	Tx ÖUT
15	CÖL2	31	STB2	47	PTT	63	MUTE
16	CÖL3	32		48	F LÖCK	64	CFILTE

5. PARTS PRICE LIST

Parts price is subject to change without notice.

(MODEL)	UCA PARTS PRICE LIST	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	
HR-2510		SSCW133005N	SCREW:FLAT HD +	4	\$ 0.09	C015	BCKG814720Z	CAPACITOR: CERAMIC	1	\$ 0.15
		SSCW192012N	SCREW: BIND HD +	2	\$ 0.09	C016	BCKDB11026Z	CAPACITOR: CERAMIC	1	\$ 0.14
		SSCW193006N	SCREW: BIND HD +	4	\$ 0.09	C017	BCGC511035Z	CAPACITOR: SEMI CONDUCTOR(SR)	1	\$ 0.23
		SSCW193008B	SCREW: BIND HD +	4	\$ 0.09	C018	BCCL311000Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33
		SSCW193008N	SCREW: BIND HD +	1	\$ 0.09	C019	BCGC511035Z	CAPACITOR: SEMI CONDUCTOR(SR)	1	\$ 0.23
		SSCW293508N	SCREW: TAPPING ROUND HD +	2	\$ 0.03	C020	BCGC514735Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.35
		SSCW295010N	SCREW: TAPPING ROUND HD +	4	\$ 0.05	C021			1	\$ 0.35
		SSCW295010N	SCREW: TAPPING ROUND HD +	4	\$ 0.05	C022			1	\$ 0.35
		SSCW343006B	SCREW: TAP TIGHT BIND HD +	8	\$ 0.03	C023			1	\$ 0.14
		SSCW343006N	SCREW: TAP TIGHT BIND HD +	8	\$ 0.03	C024	8CCG811505Z	CAPACITOR: CERAMIC	1	\$ 0.14
		SSCW4340030N	HEX NUT	4	\$ 0.07	C025	8CCG815091Z	CAPACITOR: CERAMIC	1	\$ 0.14
		SSCW4340030N	HEX NUT	4	\$ 0.07	C026	8CCG811015Z	CAPACITOR: CERAMIC	1	\$ 0.14
		SSCW480030Z	NUT: FLANGE	4	\$ 0.07	C027	8CEL811090Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33
		SSCW490050N	WASHER FLAT	4	\$ 0.03	C028	8CCG812205Z	CAPACITOR: CERAMIC	1	\$ 0.11
	SSCW530035N	WASHER-LOCK	2	\$ 0.02	C029	8CCG815605Z	CAPACITOR: CERAMIC	1	\$ 0.14	
	SSCW540050N	WASHER-STAR	4	\$ 0.09	C030	8CCG811030Z	CAPACITOR: CERAMIC	1	\$ 0.16	
	SSCW803008N	SCREW: P TIGHT BIND HD +	3	\$ 0.02	C031	8CCG511025Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.21	
	SSCW133005N	SCREW: FLAT HD +	4	\$ 0.09	C032	8CEL814790Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
	SSCW193006N	SCREW: BIND HD +	2	\$ 0.09	C033	8CEL114700Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.34	
	SSCW4803008N	SCREW: P TIGHT BIND HD +	1	\$ 0.02	C034			1	\$ 0.34	
	SSCW193006N	SCREW: BIND HD +	4	\$ 0.09	C035	8CGC513935Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.35	
	SSCW4803008N	SCREW: P TIGHT BIND HD +	2	\$ 0.02	C036	8CGC516825Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.21	
			4	\$ 0.03	C037	8CCG811804Z	CAPACITOR: CERAMIC	1	\$ 0.17	
			4	\$ 0.09	C038	8CKB811025Z	CAPACITOR: CERAMIC	1	\$ 0.16	
			2	\$ 0.09	C040	8CEL114700Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.34	
			1	\$ 0.02	C041	8CEL114706Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.34	
			4	\$ 0.09	C042	8CEL114700Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.34	
			2	\$ 0.02	C043	8CCU815091Z	CAPACITOR: CERAMIC	1	\$ 0.17	
			1	\$ 0.02	C044	8CCG811015Z	CAPACITOR: CERAMIC	1	\$ 0.23	
			1	\$ 9.85	C045	8CCF811091Z	CAPACITOR: CERAMIC	1	\$ 0.17	
			1	\$ 0.19	C046	8CKC511040Z	CAPACITOR: CERAMIC	1	\$ 0.40	
			1	\$ 1.36	C047	8CKL311000Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.12	C048	8CKG814720Z	CAPACITOR: CERAMIC	1	\$ 0.15	
			1	\$ 0.12	C049			1	\$ 0.15	
			1	\$ 8.47	C050	8CKG811030Z	CAPACITOR: CERAMIC	1	\$ 0.16	
			1	\$ 1.73	C051			1	\$ 0.16	
			1	\$ 1.06	C052	8CEL114700Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.34	
			1	\$ 0.45	C053	8CCG813304Z	CAPACITOR: CERAMIC	1	\$ 0.17	
			1	\$ 0.75	C054	8CCG811815Z	CAPACITOR: CERAMIC	1	\$ 0.16	
			1	\$ 0.75	C055	8CKG811030Z	CAPACITOR: CERAMIC	1	\$ 0.16	
			1	\$ 0.75	C056	8CE8811096Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.40	
			1	\$ 0.11	C057	8CGC516835Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.44	
			1	\$ 0.19	C058	8CGC514725Z	CAPACITOR: SEMI-CONDUCTOR(SR)	1	\$ 0.21	
			1	\$ 0.35	C059	8CEL811090Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.17	C060	8CE8112216Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.37	
			1	\$ 0.28	C061	8CCG815615Z	CAPACITOR: CERAMIC	1	\$ 0.34	
			1	\$ 0.14	C062	8CEL814780Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.14	C063	8CEL812280Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.16	C064	8CAZ111016Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.33	C065	8CEL812290Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.16	C066	8CEL811090Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	
			1	\$ 0.16	C067			1	\$ 0.33	
			1	\$ 0.16	C068	8CF8811096Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.40	
			1	\$ 0.14	C069			1	\$ 0.40	
			1	\$ 0.14	C070			1	\$ 0.40	
			1	\$ 0.14	C071	8CEL811090Z	CAPACITOR: ELECTROLYTIC	1	\$ 0.33	

(SYMBOL){PARTS CODE}	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL){PARTS CODE}	(DESCRIPTION)	(QTY)	(LIST PRICE)
C072	BGGS14725Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.0047UF 25V K	C127	BGCG813315Z	CAPACITOR:CERAMIC	330PF 50V K SL
C073	BGGS11035Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.01UF 25V K	C128	BCER812286Z	CAPACITOR:ELECTROL YTIC	0.22UF 50V M C-095
C074	BGGS14725Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.0047UF 25V K	C129	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C075	BCER112216Z	CAPACITOR:ELECTROL YTIC	220UF 10V M C-095	C130	BCER114706Z	CAPACITOR:ELECTROL YTIC	47UF 10V M C-095
C076	BCFC81091Z	CAPACITOR:CERAMIC	1PF 50V C CK	C131	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C077	BCC815091Z	CAPACITOR:CERAMIC	5PF 50V C CH	C132	BCCR812215Z	CAPACITOR:CERAMIC	220PF 50V K RH
C078	BCEL811090Z	CAPACITOR:ELECTROL YTIC	1UF 50V	C133	BGGS12235Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.022UF 25V K
C079	BCCG814715Z	CAPACITOR:CERAMIC	470PF 50V K SL	C134	BCK8815615Z	CAPACITOR:CERAMIC	560PF 50V K YB(B)
C080	BCAZ111016Z	CAPACITOR:ELECTROL YTIC	100UF 10V M C-156	C135	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C081	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF	C136	BCKD811026Z	CAPACITOR:CERAMIC	0.001UF 50V M YD
C082	BGCG811515Z	CAPACITOR:CERAMIC	150PF 50V K SL	C137	BCER112206Z	CAPACITOR:ELECTROL YTIC	22UF 10V M C-095
C083	BGCG812715Z	CAPACITOR:CERAMIC	270PF 50V K SL	C138	BKCG814720Z	CAPACITOR:CERAMIC	0.0047UF 50V Z YF
C084	BCCR812204Z	CAPACITOR:CERAMIC	22PF 50V J RH	C139	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C085	BCCR815605Z	CAPACITOR:CERAMIC	56PF 50V K RH	C140	BGGS14735Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.047UF 25V K
C086	BCCR811505Z	CAPACITOR:CERAMIC	15PF 50V K RH	C141	BCCG814715Z	CAPACITOR:CERAMIC	470PF 50V K SL
C087	BCEL814790Z	CAPACITOR:ELECTROL YTIC	4.7UF 50V	C142	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C088	BCEL114700Z	CAPACITOR:ELECTROL YTIC	47UF 10V	C143	BCCU811015Z	CAPACITOR:CERAMIC	100PF 50V K UJ
C089	BCEL811090Z	CAPACITOR:ELECTROL YTIC	1UF 50V	C144	BCCU813315Z	CAPACITOR:CERAMIC	330PF 50V K UJ
C090	BCEL114700Z	CAPACITOR:ELECTROL YTIC	47UF 10V	C145	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C091	BGGS12235Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.022UF 25V K	C146	BGGS11025Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.001UF 25V K
C092	BGGS11835Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.018UF 25V K	C147	BGGS13335Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.033UF 25V K
C093				C148	BCEL812280Z	CAPACITOR:ELECTROL YTIC	0.22UF 50V
C094	BGGS11035Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.01UF 25V K	C149	BCER311006Z	CAPACITOR:ELECTROL YTIC	10UF 16V M C-095
C095	BCER114716Z	CAPACITOR:ELECTROL YTIC	470UF 10V M C-095	C150	BKCG814730Z	CAPACITOR:CERAMIC	0.047UF 25V Z ZF
C096	BGGS11045Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.1UF 25V K	C151	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF
C097	BCER114706Z	CAPACITOR:ELECTROL YTIC	47UF 10V M C-095	C152	BCER11026Z	CAPACITOR:ELECTROL YTIC	1000UF 25V M C-095
C098	BCEL812290Z	CAPACITOR:ELECTROL YTIC	2.2UF 50V	C153	BKAZ11016Z	CAPACITOR:ELECTROL YTIC	100UF 10V M C-156
C099	BCER814796Z	CAPACITOR:ELECTROL YTIC	4.7UF 50V M C-095	C154	BCER11026Z	CAPACITOR:ELECTROL YTIC	1000UF 25V M C-095
C100	BCCG814715Z	CAPACITOR:CERAMIC	470PF 50V K SL	C155	BGGS11035Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.01UF 25V K
C101	BCER814796Z	CAPACITOR:ELECTROL YTIC	4.7UF 50V M C-095	C156	BKCG811035Z	CAPACITOR:CERAMIC	0.01UF 50V K YB(B)
C102	BGGS11045Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.1UF 25V K	C157	BCCR811015Z	CAPACITOR:CERAMIC	100PF 50V K RH
C103	BCKD811026Z	CAPACITOR:CERAMIC	0.001UF 50V M YD	C159	BCER114706Z	CAPACITOR:ELECTROL YTIC	47UF 10V M C-095
C104	BGGS14725Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.0047UF 25V K	C201	BCVR314726Z	C-CERAMIC MELF	4700PF 16V M X C-140 TAPE
C105	BGGS11035Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.01UF 25V K	C202	BKXEB14735Z	C-CERAMIC M/L (3216) TAPE	0.047UF 50V K C(B)
C106				C203	BCEL311000Z	CAPACITOR:ELECTROL YTIC	10UF 16V
C107	BGGS14735Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.047UF 25V K	C204	BCVL811004Z	C-CERAMIC MELF	10PF 50V J SL C-140 TAPE
C108	BCEL114700Z	CAPACITOR:ELECTROL YTIC	47UF 10V	C205	BKXEB14735Z	C-CERAMIC M/L (3216) TAPE	0.047UF 50V K C(B)
C109	BCEL811090Z	CAPACITOR:ELECTROL YTIC	1UF 50V	C206	BCVR314726Z	C-CERAMIC MELF	4700PF 16V M X C-140 TAPE
C110	BCER311006Z	CAPACITOR:ELECTROL YTIC	10UF 16V M C-095	C207	BCVP811025Z	C-CERAMIC MELF	0.001UF 50V K B C-140 TAPE
C111	BCEL311000Z	CAPACITOR:ELECTROL YTIC	10UF 16V	C208	BCVP818205Z	C-CERAMIC MELF	82PF 50V K B C-140 TAPE
C112	BCCR811015Z	CAPACITOR:CERAMIC	100PF 50V K RH	C209	BCVQ311036Z	C-CERAMIC MELF	0.01UF 16V M Y C-140 TAPE
C113	BCCR813305Z	CAPACITOR:CERAMIC	33PF 50V K RH	C210	BCER814786Z	CAPACITOR:ELECTROL YTIC	0.47UF 50V M C-095
C114	BCCR811815Z	CAPACITOR:CERAMIC	180PF 50V K RH	C212	BCVP813315Z	C-CERAMIC MELF	330PF 50V K B C-140 TAPE
C115				C213	BCVP811025Z	C-CERAMIC MELF	0.001UF 50V K B C-140 TAPE
C116	BCCR811515Z	CAPACITOR:CERAMIC	150PF 50V K RH	C214	BCEL311000Z	CAPACITOR:ELECTROL YTIC	10UF 16V
C117	BCCG813915Z	CAPACITOR:CERAMIC	390PF 50V K SL	C215	BCVP813315Z	C-CERAMIC MELF	330PF 50V K B C-140 TAPE
C118	BKCG814720Z	CAPACITOR:CERAMIC	0.0047UF 50V Z YF	C216	BCVQ311036Z	C-CERAMIC MELF	0.01UF 16V M Y C-140 TAPE
C119	BCER814786Z	CAPACITOR:ELECTROL YTIC	0.47UF 50V M C-095	C300	BCEL814790Z	CAPACITOR:ELECTROL YTIC	4.7UF 50V
C120	BCCG815615Z	CAPACITOR:CERAMIC	560PF 50V K SL	C301	BKXEB14735Z	C-CERAMIC M/L (3216) TAPE	0.047UF 50V K C(B)
C121	BGGS11025Z	CAPACITOR:SEMI-CONDUCTOR(SR)	0.001UF 25V K	C302	BCEL112200Z	CAPACITOR:ELECTROL YTIC	22UF 10V
C122	BKCG811030Z	CAPACITOR:CERAMIC	0.01UF 50V Z YF	C303	BCS8114796Z	CAPACITOR:TANTALUM	4.7UF 10V M C-089
C123	BCKD811026Z	CAPACITOR:CERAMIC	0.001UF 50V M YD	C304	BCVP811025Z	C-CERAMIC MELF	0.001UF 50V K B C-140 TAPE
C124	BCCG815615Z	CAPACITOR:CERAMIC	560PF 50V K SL	C305	BCVQ311036Z	C-CERAMIC MELF	0.01UF 16V M Y C-140 TAPE
C125				C306	BCEL811090Z	CAPACITOR:ELECTROL YTIC	1UF 50V
C126	BCCG818215Z	CAPACITOR:CERAMIC	820PF 50V K SL	C307			

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(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)
D106 BDAY0258001	DIODE	1	\$ 0.09	D164 BDFY0058001	VARISTOR	1	\$ 1.66
D107		1	\$ 0.09	D165		1	\$ 1.66
D108		1	\$ 0.09	D166 BDAY0258001	DIODE	1	\$ 0.09
D109		1	\$ 0.09	D167		1	\$ 0.09
D110		1	\$ 0.09	D168		1	\$ 0.09
D111 BDAY0001001	DIODE	1	\$ 0.33	D169		1	\$ 0.09
D112		1	\$ 0.33	D170		1	\$ 0.09
D113 BDAY0258001	DIODE	1	\$ 0.09	D171		1	\$ 0.09
D114		1	\$ 0.09	D172		1	\$ 0.09
D115		1	\$ 0.09	D173		1	\$ 0.09
D116		1	\$ 0.09	D174		1	\$ 0.09
D117 BDAY0001001	DIODE	1	\$ 0.33	D175		1	\$ 0.09
D118		1	\$ 0.33	D176		1	\$ 0.09
D119 BDAY0258001	DIODE	1	\$ 0.09	D177		1	\$ 0.09
D120		1	\$ 0.09	D178		1	\$ 0.09
D121		1	\$ 0.09	D201	BDAY0001001 DIODE	1	\$ 0.33
D122		1	\$ 0.09	D202		1	\$ 0.33
D123		1	\$ 0.09	D203	BDAY0433001 DIODE	1	\$ 0.24
D124		1	\$ 0.09	D301	BDAY0465001 DIODE	1	\$ 0.96
D125		1	\$ 0.09	D303		1	\$ 0.96
D126		1	\$ 0.09	D304		1	\$ 0.96
D127		1	\$ 0.09	D305		1	\$ 0.96
D128		1	\$ 0.09	D306		1	\$ 0.96
D129		1	\$ 0.09	D307		1	\$ 0.96
D131		1	\$ 0.09	D308	BDAY0281001 DIODE	1	\$ 3.77
D132		1	\$ 0.09	D311	BDAY0433001 DIODE	1	\$ 0.24
D133		1	\$ 0.09	D312	BDAY0133001 DIODE	1	\$ 0.21
D134		1	\$ 0.09	D313		1	\$ 0.21
D135		1	\$ 1.10	D315	BDAY0433001 DIODE	1	\$ 0.24
D136		1	\$ 0.09	D316	BDAY0133001 DIODE	1	\$ 0.21
D137		1	\$ 0.09	D318	BDAY0433001 DIODE	1	\$ 0.24
D138		1	\$ 0.09	D319		1	\$ 0.24
D139		1	\$ 0.09	D321		1	\$ 0.24
D140		1	\$ 0.09	D322		1	\$ 0.24
D141		1	\$ 0.09	D323		1	\$ 0.24
D142		1	\$ 0.09	D324		1	\$ 0.24
D143		1	\$ 0.09	D551	BDAY0001002 DIODE	1	\$ 0.34
D144		1	\$ 0.09	D552		1	\$ 0.34
D145		1	\$ 0.09	D552		1	\$ 0.34
D146 BDAY0269003	DIODE:ZENER	1	\$ 0.09	J301	BJKY0276003 JACK	1	\$ 0.61
D147 BDAY0269002	DIODE:ZENER	1	\$ 0.26	J302	BJKY0276004 JACK	1	\$ 0.93
D148 BDAY0258001	DIODE	1	\$ 0.26	J303		1	\$ 0.93
D149		1	\$ 0.09	J304	BJKY0276009 JACK	1	\$ 1.22
D150		1	\$ 0.09	J305		1	\$ 1.22
D151		1	\$ 0.09	J306	BJKY0276004 JACK	1	\$ 0.93
D152		1	\$ 0.09	J307	BJKY0276003 JACK	1	\$ 0.61
D153		1	\$ 0.12	J308		1	\$ 0.61
D154		1	\$ 0.12	J311	BJKY0326001 JACK	1	\$ 1.29
D155		1	\$ 0.09	J312		1	\$ 1.29
D156		1	\$ 0.09	J513	BJKY0426001 JACK	1	\$ 2.11
D157		1	\$ 0.09	J651	BJKY0191001 JACK	1	\$ 4.86
D158		1	\$ 0.09	L101	BLBY0623001 COIL	1	\$ 1.77
D159		1	\$ 0.09	L102	BLZY0051229 INDUCTOR:MOLDED	1	\$ 0.34
D160		1	\$ 0.09	L103		1	\$ 0.34
D163		1	\$ 0.09	L104	BLBY0341001 COIL	1	\$ 1.33
		1	\$ 0.09	L105	BLBY0342001 COIL	1	\$ 1.33

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY) (LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY) (LIST PRICE)
L106 BLZY0016471	INDUCTOR:MOLDED	LZ-016 5L0609-471K 470UH 1 \$ 0.46	Q112	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53
L107			Q113		
L108			Q114		
L109 BLZY0016101	INDUCTOR:MOLDED	LZ-016 5L0609-101K 100UH 1 \$ 0.46	Q115		
L111 BL8Y0342001	COIL	LB-342 199CC-13524R 1 \$ 1.33	Q116		
L113 BL8Y0341001	COIL	LB-341 199CC-13499A 1 \$ 1.33	Q117		
L115			Q118		
L116 BL8Y0209001	COIL	LB-209 113CN-6485Z 1 \$ 2.44	Q119	TRANSISTOR	DB-259 25C1675-L 1 \$ 0.61
L117 BL8Y0137001	COIL	LB-137 113CH-6344Z 1 \$ 1.06	Q120	TRANSISTOR	DB-003 25A733A-P 1 \$ 0.57
L118			Q121	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53
L121 BL8Y0096001	COIL	LE-096 8 1/2T 1 \$ 1.06	Q122		
L122 BL8Y0092001	COIL	LE-092 6 1/2T 1 \$ 0.17	Q123		
L123 BL8Y0093001	COIL	LE-093 7 1/2T 1 \$ 0.16	Q124	TRANSISTOR	DB-272 25C1973-SSB 1 \$ 3.25
L124 BLDY0228001	COIL	LD-228 A1050T-3012 1 \$ 0.70	Q125	TRANSISTOR	DB-106 25B825-C 1 \$ 0.75
L126 BLDY0221001	COIL	LD-221 1 \$ 1.62	Q126	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53
L129 BLDY0087001	COIL	LD-087 BF04-3*5*1 1 \$ 0.20	Q127	TRANSISTOR	DB-383 25C3242A-E 1 \$ 0.81
L131			Q128	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53
L132 BL8Y0201001	COIL	LE-201 D2.4 3 1/2T 1 \$ 0.19	Q129		
L134 BLDY0229001	COIL	LD-229 1 \$ 3.07	Q131		
L135 BLDY0230001	COIL	LD-230 1 \$ 3.89	Q132	TRANSISTOR	DB-529 MRF-477 1 \$ 0.53
L136			Q133	TRANSISTOR	DB-331 25C2166-C 1 \$ 5.02
L203 BLFY0125001	COIL	LF-125 332PC-2673Y 1 \$ 2.51	Q134	TRANSISTOR	DB-228 25C2086-D 1 \$ 2.19
L301 BL8Y0246001	COIL	LF-246 D0.6 3 1/2T 1 \$ 1.87	Q135	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53
L302 BLFY0132001	COIL	LF-132 A2945N-14641B 1 \$ 1.87	Q136		
L303 BLFY0131001	COIL	LF-131 A2945N-1373AQ 1 \$ 1.87	Q137	TRANSISTOR	DB-381 25C2712-Y TAPING 1 \$ 0.42
L304			Q138	TRANSISTOR	DB-003 25A733A-P 1 \$ 0.57
L305 BLZY0051339	INDUCTOR:MOLDED	LZ-051 5P0305-3R3K 3.3UH 1 \$ 0.81	Q201	TRANSISTOR	DB-744 25C2814-F5 TAPING 1 \$ 0.40
L306 BLZY0051479	INDUCTOR:MOLDED	LZ-051 5P0305-4R7K 4.7UH 1 \$ 1.06	Q202		
L307			Q203		
L308 BLZY0051100	INDUCTOR:MOLDED	LZ-051 5P0305-100K 100H 1 \$ 0.81	Q204	TRANSISTOR	DB-743 25C2812-L5 TAPING 1 \$ 0.40
L309 BLZY0051689	INDUCTOR:MOLDED	LZ-051 5P0305-6R8K 6.8UH 1 \$ 0.34	Q205		
L310 BLFY0159001	COIL	LF-159 341LC-1286N 1 \$ 2.51	Q206	TRANSISTOR	DB-048 25A1179-M6 TAPING 1 \$ 0.36
L311 BLZY0051689	INDUCTOR:MOLDED	LZ-051 5P0305-6R8K 6.8UH 1 \$ 0.34	Q207	TRANSISTOR	DB-744 25C2814-F5 TAPING 1 \$ 0.40
L312 BLZY0051279	INDUCTOR:MOLDED	LZ-051 5P0305-2R7M 2.7UH 1 \$ 0.34	Q301	TRANSISTOR	DB-724 25C3121 TAPING 1 \$ 1.14
L313 BLFY0130001	COIL	LF-130 A2945N-1372AQ 1 \$ 1.87	Q302		
L314 BLFY0131001	COIL	LF-131 A2945N-1373AQ 1 \$ 1.87	Q303		
L315 BLFY0172001	COIL	LF-172 A2945N-15291B 1 \$ 1.87	Q304		
L316 BLZY0044682	INDUCTOR:MOLDED	LZ-044 262LY-682K 6.8MH 1 \$ 1.50	Q305	FIELD EFFECT TRANSISTOR	DC-025 25K302-Y TAPING 1 \$ 1.45
L317 BLFY0134001	COIL	LF-134 A2945NS-15071B 1 \$ 1.87	Q306		
L318 BLFY0133001	COIL	LF-133 A2945NS-14411B 1 \$ 1.87	Q307		
L319 BLFY0158001	COIL	LF-158 341LC-1282EQ 1 \$ 2.51	Q308	TRANSISTOR	DB-744 25C2814-F5 TAPING 1 \$ 0.40
L401 BLFY0072001	COIL	LF-072 SPLC-1055Z 1 \$ 2.76	Q311	TRANSISTOR	DB-048 25A1179-M6 TAPING 1 \$ 0.36
L651 BLZY0051471	INDUCTOR:MOLDED	LZ-051 5P0305-471K 470UH 1 \$ 0.81	Q312		
L652			Q313		
L653			Q314		
Q101 B08C1674111	TRANSISTOR	DB-295 25C1674-L 1 \$ 0.73	Q315	TRANSISTOR	DB-381 25C2712-Y TAPING 1 \$ 0.42
Q102 B0C80192533	FIELD EFFECT TRANSISTOR	DC-019 25K192A-BL 1 \$ 0.99	Q316	TRANSISTOR	DB-048 25A1179-M6 TAPING 1 \$ 0.36
Q103 B08C1675111	TRANSISTOR	DB-259 25C1675-L 1 \$ 0.61	Q317	TRANSISTOR	DB-381 25C2712-Y TAPING 1 \$ 0.42
Q104 B08C1674111	TRANSISTOR	DB-295 25C1674-L 1 \$ 0.73	Q401	TRANSISTOR	DB-744 25C2814-F5 TAPING 1 \$ 0.40
Q105 B08C1675111	TRANSISTOR	DB-259 25C1675-L 1 \$ 0.61	R001	R:CARBON AXIAL LEAD(TAPE)	5.6K 1/6W J 1 \$ 0.05
Q106			R002	R:CARBON AXIAL LEAD(TAPE)	8.2K 1/6W J 1 \$ 0.05
Q107 B08C1730111	TRANSISTOR	DB-269 25C1730-L 1 \$ 1.21	R003	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J 1 \$ 0.05
Q108 B08C0945507	TRANSISTOR	DB-224 25C945A-Q 1 \$ 0.53	R004	R:CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J 1 \$ 0.05
Q109			R005	RESISTOR:CARBON MELF CHIP	68K 1/8W J TAPING 1 \$ 0.09
Q111			R006	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J 1 \$ 0.05

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)
R007	BRFT184724Z	R:CARBON AXIAL LEAD(TAPE)	4.7K 1/8W J	R063	BRFT614734Z	R:CARBON AXIAL LEAD(TAPE)	47K 1/6W J
R008	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J	R064	BRFT613324Z	R:CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J
R009	BRFT613324Z	R:CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J	R065	DRUB612224Z	RESISTOR:CARBON FORMED VERT	2.2K 1/6W J
R010	BRFT613314Z	R:CARBON AXIAL LEAD(TAPE)	330 1/6W J	R066	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J
R011	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J	R067	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J
R012	BRFT618214Z	R:CARBON AXIAL LEAD(TAPE)	820 1/6W J	R068	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J
R013	BRFT616814Z	R:CARBON AXIAL LEAD(TAPE)	680 1/6W J	R069	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J
R014	BRFT618224Z	R:CARBON AXIAL LEAD(TAPE)	8.2K 1/6W J	R070	DRFT613924Z	R:CARBON AXIAL LEAD(TAPE)	3.9K 1/6W J
R015	BRFT612234Z	R:CARBON AXIAL LEAD(TAPE)	22K 1/6W J	R071	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J
R016	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J	R072	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J
R018	BRFT615624Z	R:CARBON AXIAL LEAD(TAPE)	5.6K 1/6W J	R073	BRFT613314Z	R:CARBON AXIAL LEAD(TAPE)	330 1/6W J
R019	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J	R074	BRFT611514Z	R:CARBON AXIAL LEAD(TAPE)	150 1/6W J
R020	BRFT615624Z	R:CARBON AXIAL LEAD(TAPE)	5.6K 1/6W J	R075	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J
R021	BRFT611224Z	R:CARBON AXIAL LEAD(TAPE)	1.2K 1/6W J	R076	BRFT611544Z	R:CARBON AXIAL LEAD(TAPE)	150K 1/6W J
R022	BRFD182234Z	RESISTOR:CARBON MELF CHIP	22K 1/8W J	R077	BRFT612744Z	R:CARBON AXIAL LEAD(TAPE)	270K 1/6W J
R023	BRFT616814Z	R:CARBON AXIAL LEAD(TAPE)	680 1/6W J	R078	BRFT611554Z	R:CARBON AXIAL LEAD(TAPE)	1.5M 1/6W J
R024	BRUB612224Z	RESISTOR:CARBON FORMED VERT	2.2K 1/6W J	R079	BRUB618224Z	RESISTOR:CARBON FORMED VERT	8.2K 1/6W J
R026	BRFT614734Z	R:CARBON AXIAL LEAD(TAPE)	47K 1/6W J	R080	BRUB611044Z	RESISTOR:CARBON FORMED VERT	100K 1/6W J
R027	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J	R081	BRFT614734Z	R:CARBON AXIAL LEAD(TAPE)	47K 1/6W J
R028	BRFT612714Z	R:CARBON AXIAL LEAD(TAPE)	270 1/6W J	R082	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J
R029	BRFT611514Z	R:CARBON AXIAL LEAD(TAPE)	150 1/6W J	R084	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J
R030	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J	R085	BRFT614744Z	R:CARBON AXIAL LEAD(TAPE)	470K 1/6W J
R031	BRUB616804Z	RESISTOR:CARBON FORMED VERT	68 1/6W J	R086	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J
R032	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J	R087	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J
R033	BRFT613324Z	R:CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J	R088	BRFT612734Z	R:CARBON AXIAL LEAD(TAPE)	27K 1/6W J
R034	BRUB611524Z	RESISTOR:CARBON FORMED VERT	1.5K 1/6W J	R089	BRFT614734Z	R:CARBON AXIAL LEAD(TAPE)	47K 1/6W J
R035	BRFT611044Z	R:CARBON AXIAL LEAD(TAPE)	100K 1/6W J	R090	BRFT612734Z	R:CARBON AXIAL LEAD(TAPE)	27K 1/6W J
R036	BRUB611044Z	RESISTOR:CARBON FORMED VERT	100K 1/6W J	R091	BRUB614734Z	RESISTOR:CARBON FORMED VERT	47K 1/6W J
R038	BRFT611044Z	R:CARBON AXIAL LEAD(TAPE)	100K 1/6W J	R092	BRUB615604Z	RESISTOR:CARBON FORMED VERT	56 1/6W J
R039	BRUB616814Z	RESISTOR:CARBON FORMED VERT	680 1/6W J	R093	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J
R040	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J	R094	BRUB614724Z	RESISTOR:CARBON FORMED VERT	4.7K 1/6W J
R041	BRFT613334Z	R:CARBON AXIAL LEAD(TAPE)	33K 1/6W J	R095			
R042	BRFT618234Z	R:CARBON AXIAL LEAD(TAPE)	82K 1/6W J	R096	BRUB612754Z	RESISTOR:CARBON FORMED VERT	2.7M 1/6W J
R043	BRFT613934Z	R:CARBON AXIAL LEAD(TAPE)	39K 1/6W J	R097	BRUB613324Z	RESISTOR:CARBON FORMED VERT	3.3K 1/6W J
R044	BRFT618244Z	R:CARBON AXIAL LEAD(TAPE)	820K 1/6W J	R098	BRUB614724Z	RESISTOR:CARBON FORMED VERT	4.7K 1/6W J
R045	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J	R099			
R046	BRFT611544Z	R:CARBON AXIAL LEAD(TAPE)	150K 1/6W J	R100	BRFT613324Z	R:CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J
R047	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J	R101	BRUB612244Z	RESISTOR:CARBON FORMED VERT	220K 1/6W J
R048	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J	R102	BRFT611524Z	R:CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J
R049	BRUB611524Z	RESISTOR:CARBON FORMED VERT	1.5K 1/6W J	R103	BRFT614734Z	R:CARBON AXIAL LEAD(TAPE)	47K 1/6W J
R050	BRFT612744Z	R:CARBON AXIAL LEAD(TAPE)	270K 1/6W J	R104	BRUB614744Z	RESISTOR:CARBON FORMED VERT	470K 1/6W J
R051				R105	BRUB614714Z	RESISTOR:CARBON FORMED VERT	470 1/6W J
R052	BRFT611044Z	R:CARBON AXIAL LEAD(TAPE)	100K 1/6W J	R106	BRUB615624Z	RESISTOR:CARBON FORMED VERT	5.6K 1/6W J
R053	BRFT611034Z	R:CARBON AXIAL LEAD(TAPE)	10K 1/6W J	R107	BRUB612234Z	RESISTOR:CARBON FORMED VERT	22K 1/6W J
R054	BRUB611044Z	RESISTOR:CARBON FORMED VERT	100K 1/6W J	R108	BRUB614714Z	RESISTOR:CARBON FORMED VERT	470 1/6W J
R055	BRFT611044Z	R:CARBON AXIAL LEAD(TAPE)	100K 1/6W J	R109	BRUB611524Z	RESISTOR:CARBON FORMED VERT	1.5K 1/6W J
R056	BRUB616834Z	RESISTOR:CARBON FORMED VERT	68K 1/6W J	R110	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J
R057	BRFT611054Z	R:CARBON AXIAL LEAD(TAPE)	1M 1/6W J	R111			
R058	BRFT616834Z	R:CARBON AXIAL LEAD(TAPE)	68K 1/6W J	R112	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J
R059	BRUB612244Z	RESISTOR:CARBON FORMED VERT	220K 1/6W J	R113	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J
R060	BRUB612734Z	RESISTOR:CARBON FORMED VERT	27K 1/6W J	R114	BRFT612224Z	R:CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J
R061	BRFT183324Z	R:CARBON AXIAL LEAD(TAPE)	3.3K 1/8W J	R115	BRFT611014Z	R:CARBON AXIAL LEAD(TAPE)	100 1/6W J
R062	BRFT611024Z	R:CARBON AXIAL LEAD(TAPE)	1K 1/6W J	R116	BRFT614714Z	R:CARBON AXIAL LEAD(TAPE)	470 1/6W J
				R117	BRFT611834Z	R:CARBON AXIAL LEAD(TAPE)	18K 1/6W J

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)			
R118	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	1	\$	0.05	1	\$	0.05	
R119	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J	1	\$	0.07	68K 1/6W J	1	\$	0.05
R120	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	1	\$	0.05	33 1/6W J	1	\$	0.05
R121	BRUB613324Z	RESISTOR:CARBON FORMED VERT	3.3K 1/6W J	1	\$	0.07	100 1/6W J	1	\$	0.05
R122	BRFT613324Z	R-CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J	1	\$	0.05	1.5K 1/6W J	1	\$	0.05
R123	BRUB618214Z	RESISTOR:CARBON FORMED VERT	820 1/6W J	1	\$	0.07	47K 1/6W J	1	\$	0.05
R124	BRFT611024Z	R-CARBON AXIAL LEAD(TAPE)	1K 1/6W J	1	\$	0.05	3.3K 1/6W J	1	\$	0.05
R125	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J	1	\$	0.07	47 1/6W J	1	\$	0.05
R126	BRUB612234Z	RESISTOR:CARBON FORMED VERT	22K 1/6W J	1	\$	0.07	1.5K 1/6W J	1	\$	0.05
R127	BRUB611524Z	RESISTOR:CARBON FORMED VERT	1.5K 1/6W J	1	\$	0.07	470 1/6W J	1	\$	0.05
R128	BRFT611024Z	R-CARBON AXIAL LEAD(TAPE)	1K 1/6W J	1	\$	0.05	2.2K 1/6W J	1	\$	0.05
R129	BRUB611024Z	RESISTOR:CARBON FORMED VERT	1K 1/6W J	1	\$	0.07	10K 1/6W J	1	\$	0.05
R130	BRFT612224Z	R-CARBON AXIAL LEAD(TAPE)	2.2K 1/6W J	1	\$	0.05	1.5K 1/6W J	1	\$	0.05
R131	BRFT611034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/6W J	1	\$	0.05	47K 1/6W J	1	\$	0.05
R132	BRFT614744Z	R-CARBON AXIAL LEAD(TAPE)	470K 1/6W J	1	\$	0.05	10K 1/6W J	1	\$	0.05
R133	BRFT614724Z	R-CARBON AXIAL LEAD(TAPE)	4.7K 1/6W J	1	\$	0.05	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R134	BRUB614724Z	RESISTOR:CARBON FORMED VERT	4.7K 1/6W J	1	\$	0.07	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R135	BRUB612244Z	RESISTOR:CARBON FORMED VERT	220K 1/6W J	1	\$	0.07	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R136	BRFT611034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/6W J	1	\$	0.05	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R137	BRUB614724Z	RESISTOR:CARBON FORMED VERT	4.7K 1/6W J	1	\$	0.07	R-CARBON AXIAL LEAD(TAPE)	1	\$	0.05
R138	BRUB612234Z	RESISTOR:CARBON FORMED VERT	22K 1/6W J	1	\$	0.07	R-CARBON AXIAL LEAD(TAPE)	1	\$	0.05
R139	BRFT611034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/6W J	1	\$	0.05	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R140	BRFT615614Z	R-CARBON AXIAL LEAD(TAPE)	560 1/6W J	1	\$	0.05	R-CARBON AXIAL LEAD(TAPE)	1	\$	0.03
R141	BRZY0025001	JUMPER CHIP	RZ-025 MRD18S TAPING	1	\$	0.09	RESISTOR:CARBON FORMED VERT	1	\$	0.03
R142	BRUB611034Z	RESISTOR:CARBON FORMED VERT	10K 1/6W J	1	\$	0.07	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R143	BRFT611094Z	R-CARBON AXIAL LEAD(TAPE)	1 1/6W J	1	\$	0.05	RESISTOR:CARBON FORMED VERT	1	\$	0.07
R144	BRFT611044Z	R-CARBON AXIAL LEAD(TAPE)	100K 1/6W J	1	\$	0.05	JUMPER CHIP	1	\$	0.09
R145	BRFT612214Z	R-CARBON AXIAL LEAD(TAPE)	220 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R146	BRFT612234Z	R-CARBON AXIAL LEAD(TAPE)	22K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R147	BRFT615634Z	R-CARBON AXIAL LEAD(TAPE)	56K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.39
R148	BRFT614724Z	R-CARBON AXIAL LEAD(TAPE)	4.7K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.39
R149	BRFT613324Z	R-CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R150	BRUB611234Z	RESISTOR:CARBON FORMED VERT	12K 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R151	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R152	BRFT615614Z	R-CARBON AXIAL LEAD(TAPE)	560 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R153	BRUB613304Z	RESISTOR:CARBON FORMED VERT	33 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R154	BRFT616844Z	R-CARBON AXIAL LEAD(TAPE)	680K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R155	BRFT611034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/6W J	1	\$	0.05	R-CARBON AXIAL LEAD(TAPE)	1	\$	0.03
R156	BRFT613324Z	R-CARBON AXIAL LEAD(TAPE)	3.3K 1/6W J	1	\$	0.05	JUMPER CHIP	1	\$	0.09
R157	BRFT611004Z	R-CARBON AXIAL LEAD(TAPE)	10 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R158	BRFT128204Z	R-CARBON AXIAL LEAD(TAPE)	82 1/2W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R159	BRFT613314Z	R-CARBON AXIAL LEAD(TAPE)	330 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R160	BRUB611024Z	RESISTOR:CARBON FORMED VERT	1K 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R161	BRUB614704Z	RESISTOR:CARBON FORMED VERT	47 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R162	BRFT611014Z	R-CARBON AXIAL LEAD(TAPE)	100 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R163	BRFT611504Z	R-CARBON AXIAL LEAD(TAPE)	15 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R164	BRFT611514Z	R-CARBON AXIAL LEAD(TAPE)	150 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R165	BRUB613314Z	RESISTOR:CARBON FORMED VERT	330 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R166	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R167	BRUB611014Z	RESISTOR:CARBON FORMED VERT	100 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R168	BRFT611024Z	R-CARBON AXIAL LEAD(TAPE)	1K 1/6W J	1	\$	0.05	R-CARBON AXIAL LEAD(TAPE)	1	\$	0.05
R169	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.05
R170	BRUB611024Z	RESISTOR:CARBON FORMED VERT	1K 1/6W J	1	\$	0.07	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R171	BRFT611014Z	R-CARBON AXIAL LEAD(TAPE)	100 1/6W J	1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09
R172				1	\$	0.05	RESISTOR:CARBON MELF CHIP	1	\$	0.09

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)
R309	BRFT183324Z	R-CARBON AXIAL LEAD(TAPE)	3.3K 1/8W J	R371	BRFD182224Z	RESISTOR:CARBON MELF CHIP	2.2K 1/8W J TAPING
R311	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING	R372	BRFD181054Z	RESISTOR:CARBON MELF CHIP	1M 1/8W J TAPING
R312				R374	BRFD181044Z	RESISTOR:CARBON MELF CHIP	100K 1/8W J TAPING
R313	BRFD181534Z	RESISTOR:CARBON MELF CHIP	15K 1/8W J TAPING	R375	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING
R314	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING	R376			
R315				R377	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING
R316	BRFD181234Z	RESISTOR:CARBON MELF CHIP	12K 1/8W J TAPING	R379	BRFD181044Z	RESISTOR:CARBON MELF CHIP	100K 1/8W J TAPING
R317	BRFT181034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/8W J	R381	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING
R318	BRFD181014Z	RESISTOR:CARBON MELF CHIP	100 1/8W J TAPING	R382			
R319				R383			
R320	BRFD183314Z	RESISTOR:CARBON MELF CHIP	330 1/8W J TAPING	R384			
R321	BRFT183314Z	R-CARBON AXIAL LEAD(TAPE)	330 1/8W J	R385	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING
R322	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING	R387			
R323	BRFT611024Z	R-CARBON AXIAL LEAD(TAPE)	1K 1/6W J	R391	BRFD182744Z	RESISTOR:CARBON MELF CHIP	270K 1/8W J TAPING
R324	BRFD184704Z	RESISTOR:CARBON MELF CHIP	47 1/8W J TAPING	R392	BRFD184724Z	RESISTOR:CARBON MELF CHIP	4.7K 1/8W J TAPING
R325	BRFD182224Z	RESISTOR:CARBON MELF CHIP	2.2K 1/8W J TAPING	R393	BRFD184714Z	RESISTOR:CARBON MELF CHIP	470 1/8W J TAPING
R326	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING	R394	BRFD186814Z	RESISTOR:CARBON MELF CHIP	680 1/8W J TAPING
R327	BRFD188214Z	RESISTOR:CARBON MELF CHIP	820 1/8W J TAPING	R395	BRFD186824Z	RESISTOR:CARBON MELF CHIP	6.8K 1/8W J TAPING
R328	BRUB184714Z	RESISTOR:CARBON FORMED VERT	470 1/8W J	R402	BRFD181234Z	RESISTOR:CARBON MELF CHIP	12K 1/8W J TAPING
R329	BRFD184704Z	RESISTOR:CARBON MELF CHIP	47 1/8W J TAPING	R403	BRFD182234Z	RESISTOR:CARBON MELF CHIP	22K 1/8W J TAPING
R331	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING	R404	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING
R332				R405			
R333	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING	R406	BRFT614714Z	R-CARBON AXIAL LEAD(TAPE)	470 1/6W J
R334				R407	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING
R335	BRFD183324Z	RESISTOR:CARBON MELF CHIP	3.3K 1/8W J TAPING	R408	BRFD181014Z	RESISTOR:CARBON MELF CHIP	100 1/8W J TAPING
R336	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING	R409	BRFD181004Z	RESISTOR:CARBON MELF CHIP	10 1/8W J TAPING
R337				R411	BRFD182224Z	RESISTOR:CARBON MELF CHIP	2.2K 1/8W J TAPING
R338	BRFD182234Z	RESISTOR:CARBON MELF CHIP	22K 1/8W J TAPING	R412			
R339	BRFD181814Z	RESISTOR:CARBON MELF CHIP	180 1/8W J TAPING	R413	BRFD184734Z	RESISTOR:CARBON MELF CHIP	47K 1/8W J TAPING
R341				R414	BRFD186834Z	RESISTOR:CARBON MELF CHIP	68K 1/8W J TAPING
R342	BRFD183324Z	RESISTOR:CARBON MELF CHIP	3.3K 1/8W J TAPING	R415	BRFD181014Z	RESISTOR:CARBON MELF CHIP	100 1/8W J TAPING
R343	BRFD182214Z	RESISTOR:CARBON MELF CHIP	220 1/8W J TAPING	R416	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING
R344	BRFD184724Z	RESISTOR:CARBON MELF CHIP	4.7K 1/8W J TAPING	R451			
R345	BRFD185624Z	RESISTOR:CARBON MELF CHIP	5.6K 1/8W J TAPING	R457			
R346	BRFD186814Z	RESISTOR:CARBON MELF CHIP	680 1/8W J TAPING	R458			
R347	BRFT611524Z	R-CARBON AXIAL LEAD(TAPE)	1.5K 1/6W J	R459			
R348	BRFD183324Z	RESISTOR:CARBON MELF CHIP	3.3K 1/8W J TAPING	R461			
R349	BRFD181814Z	RESISTOR:CARBON MELF CHIP	180 1/8W J TAPING	R462			
R351	BRFD181524Z	RESISTOR:CARBON MELF CHIP	1.5K 1/8W J TAPING	R463			
R352	BRFD181814Z	RESISTOR:CARBON MELF CHIP	180 1/8W J TAPING	R464			
R353	BRFD183324Z	RESISTOR:CARBON MELF CHIP	3.3K 1/8W J TAPING	R465			
R354	BRFT184724Z	R-CARBON AXIAL LEAD(TAPE)	4.7K 1/8W J	R466			
R355	BRZY0025001	JUMPER CHIP	RZ-025 MRO18S TAPING	R478			
R356	BRFT611034Z	R-CARBON AXIAL LEAD(TAPE)	10K 1/6W J	R481			
R357	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING	R482	BRFD184724Z	RESISTOR:CARBON MELF CHIP	4.7K 1/8W J TAPING
R358	BRFD184724Z	RESISTOR:CARBON MELF CHIP	4.7K 1/8W J TAPING	R483			
R359	BRFD181224Z	RESISTOR:CARBON MELF CHIP	1.2K 1/8W J TAPING	R484			
R360	BRFD186834Z	RESISTOR:CARBON MELF CHIP	68K 1/8W J TAPING	R485			
R361	BRFD181224Z	RESISTOR:CARBON MELF CHIP	1.2K 1/8W J TAPING	R486			
R362	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING	R487			
R363	BRFD181024Z	RESISTOR:CARBON MELF CHIP	1K 1/8W J TAPING	R488	BRFD181034Z	RESISTOR:CARBON MELF CHIP	10K 1/8W J TAPING
R364	BRFD185624Z	RESISTOR:CARBON MELF CHIP	5.6K 1/8W J TAPING	R489			
R365	BRFT613934Z	R-CARBON AXIAL LEAD(TAPE)	39K 1/6W J	R491			
R366	BRFD183324Z	RESISTOR:CARBON MELF CHIP	3.3K 1/8W J TAPING	R492			
R370	BRFD181524Z	RESISTOR:CARBON MELF CHIP	1.5K 1/8W J TAPING	R495			

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY) (LIST PRICE)	(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY) (LIST PRICE)
JP025	BYYY0052007 JUMPER WIRE	YY-052 20.0MM 1 \$ 0.03	JP801	BYTY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03
JP026	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	JP802	BYTY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03
JP027	BYYY0052005 JUMPER WIRE	YY-052 15.0MM 1 \$ 0.03	LC501	BOLY0027001 LIQUID CRYSTAL DISPLAY	IL-027 LMS34C-A 1 \$ 24.72
JP028	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	MC511	BPLY0088001 MICROPHONE	MC-388 1 \$ 24.00
JP029	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	PL501	BPLY0102001 PILOT LAMP	PL-102 1 \$ 1.99
JP030			RR301	BHAY0117001 R:ARRAY	HA-117 1 \$ 2.67
JP031			RR302	BHAY0096001 R:ARRAY	HA-096 47K*9 1 \$ 2.90
JP032	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	RR303	BHAY0093001 R:ARRAY	HA-093 1 \$ 1.73
JP033	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	SP551	8SPY0181001 SPEAKER	SP-181 1 \$ 6.24
JP034	BYYY0052007 JUMPER WIRE	YY-052 20.0MM 1 \$ 0.03	TP002	BTPY0044001 TERMINAL:CHECK POINT	TP-044 1 \$ 0.21
JP035	BYYY0052005 JUMPER WIRE	YY-052 15.0MM 1 \$ 0.03	TP003		
JP036	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	TP004		
JP037			VR101	BRTY0528223 RESISTOR:SEMI-FIXED	RT-528 220K 1 \$ 0.21
JP038	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	VR102	BRTY0528473 RESISTOR:SEMI-FIXED	RT-528 470K 1 \$ 1.06
JP039			VR103	BRTY0528103 RESISTOR:SEMI-FIXED	RT-528 100K 1 \$ 1.06
JP040	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	VR104		
JP041	BYYY0052007 JUMPER WIRE	YY-052 20.0MM 1 \$ 0.03	VR105		
JP042	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	VR106		
JP043	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	VR107	BRTY0528473 RESISTOR:SEMI-FIXED	RT-528 470K 1 \$ 1.06
JP044	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	VR111	BRTY0528103 RESISTOR:SEMI-FIXED	RT-528 100K 1 \$ 1.06
JP045	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	VR112	BRTY0182101 RESISTOR:SEMI-FIXED	RT-182 TT24R 100K 1 \$ 1.01
JP046			VR113	BRTY0182102 RESISTOR:SEMI-FIXED	RT-182 TT24R 1K 1 \$ 1.01
JP047			VR114	BRTY0528102 RESISTOR:SEMI-FIXED	RT-528 1K 1 \$ 1.06
JP048			VR115	BRTY0528103 RESISTOR:SEMI-FIXED	RT-528 100K 1 \$ 1.06
JP049	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	VR116		
JP050	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	VR117	BRTY0528473 RESISTOR:SEMI-FIXED	RT-528 470K 1 \$ 1.06
JP051			VR301	BRTY0528101 RESISTOR:SEMI-FIXED	RT-528 100K 1 \$ 1.06
JP052			VR302		
JP053			VR601	BRY0672001 RESISTOR:VARIABLE	RV-672 5K 1 \$ 5.27
JP054	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	VR701	BRY0677001 RESISTOR:VARIABLE	RV-677 RK0971111 50KA W/S 1 \$ 5.27
JP055	BYYY0052004 JUMPER WIRE	YY-052 12.5MM 1 \$ 0.03	VR702	BRY0678001 RESISTOR:VARIABLE	RV-678 RK0971111 50KB W/S 1 \$ 5.27
JP056			VR801	BRY0652001 RESISTOR:VARIABLE	RV-652 VBI2L BIK 1 \$ 5.27
JP057			VR802	BRY0679001 RESISTOR:VARIABLE	RV-679 10K 1 \$ 5.27
JP058			CZD071399Z	WIRES ASSEMBLED	W-071399 1 \$ 9.08
JP059			CZD071398Z	WIRES ASSEMBLED	W-071398 1 \$ 6.81
JP060	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	CZD071396Z	WIRES ASSEMBLED	W-071396 1 \$ 9.08
JP061	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	CZD071395Z	WIRES ASSEMBLED	W-071395 1 \$ 6.81
JP062	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	CZD071394Z	WIRES ASSEMBLED	W-071394 1 \$ 6.83
JP063	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	CZD071392Z	WIRES ASSEMBLED	W-071392 1 \$ 3.40
JP064	BYYY0052004 JUMPER WIRE	YY-052 12.5MM 1 \$ 0.03	CZD071391Z	WIRES ASSEMBLED	W-071391 1 \$ 20.40
JP065	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	CZD071389Z	WIRES ASSEMBLED	W-071389 1 \$ 2.26
JP066	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	CZD071388Z	WIRES ASSEMBLED	W-071388 1 \$ 4.54
JP067	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	WA015	BWZY0692001 CORD	WZ-692 230 W/P 1 \$ 4.05
JP068	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	WA016	BWZY0693001 CORD	WZ-693 270 W/P 1 \$ 4.19
JP069	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	WA018	BWZY0727001 CORD	WZ-727 1 \$ 0.07
JP070	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	WA019	BWZY0721001 DC CORD	WZ-721 1830 W/P 1 \$ 0.07
JP071			CZD071401Z	WIRES ASSEMBLED	W-071401 1 \$ 2.81
JP072	BYYY0052004 JUMPER WIRE	YY-052 12.5MM 1 \$ 0.03	WA021	BWZY0715001 WIRE ASSEMBLED	WZ-715 120 W/P 1 \$ 0.65
JP073			WA022	BWZY0716001 WIRE ASSEMBLED	WZ-716 70 W/P 1 \$ 0.65
JP074	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	WA023	BWZY0725001 WIRE ASSEMBLED	WZ-725 125 W/P 1 \$ 8.91
JP075	BYYY0052003 JUMPER WIRE	YY-052 10.0MM 1 \$ 0.03	WA024	BWZY0726001 WIRE ASSEMBLED	WZ-726 120 W/P 1 \$ 8.91
JP076	BYYY0052001 JUMPER WIRE	YY-052 5.0MM 1 \$ 0.03	CZD071422Z	WIRE ASSEMBLY	W-071422 1 \$ 0.65
JP077			Y1101	BYDY0010003 INSULATION SHEET	YD-010 P101KD 1 \$ 0.28
JP078	BYYY0052002 JUMPER WIRE	YY-052 7.5MM 1 \$ 0.03	Y1102	BYDY00172001 BUSHING:TIGHT CERAMIC	YY-172 1 \$ 0.28
JP079			Y1103		

(SYMBOL)(PARTS CODE)	(DESCRIPTION)	(QTY)	(LIST PRICE)
YWS01	8YYY0047001 WIRE CLAMPER	1	\$ 0.09
YWS02		1	\$ 0.09
YWS03		1	\$ 0.09
YWS04		1	\$ 0.09
YWS05		1	\$ 0.09
YWS06		1	\$ 0.09
YWS07		1	\$ 0.09
YWS08		1	\$ 0.09
YWS09		1	\$ 0.09
YWS10		1	\$ 0.09
FCR221155Z	CHASSIS:REAR	1	\$ 4.05
GCMZ420938Z	PLATE:DISPLAY	1	\$ 5.62
GHDZ320936Z	HOLDER:LCD	1	\$ 0.54
GMS480407Z	SCREW:MOUNTING	4	\$ 0.54
GNBP420940Z	BUTTON:CHANNEL	2	\$ 0.48
GNBP421146Z	KNOB:PUSH	5	\$ 0.33
GNBP421147Z	KNOB:PUSH	6	\$ 0.33
HBC7380105A	MOUNTING BRACKET	1	\$ 6.01
HCM8221190A	CASE:BOTTOM	1	\$ 6.53
HCM7221191Z	CASE:TOP	1	\$ 4.05
HCS5321192B	CHASSIS:SIDE	2	\$ 0.12
HHSK418834B	HEAT SINK	1	\$ 0.09
HHGA02919Z	HANGER:MICROPHONE	1	\$ 0.10
HSDP418267Z	SHIELD PLATE	1	\$ 0.33
HSDP421141Z	FRAME:LCD	1	\$ 0.33
HSDP421142Z	SHIELD PLATE	1	\$ 0.33
HSDP421143Z	PLATE:ESD (R)	1	\$ 0.33
HSDP421144Z	PLATE:ESD (L)	1	\$ 0.33
HWSR480823Z	WASHER:ANT	1	\$ 0.03
PLBC482147Z	LABEL:WARNING,DC CORD	1	\$ 0.15
RETC421189Z	PLATE:LCD	1	\$ 0.09
RUTC403865Z	WOOL-COATED PAPER:WOOL TACK	4	\$ 0.03
RZEB418268Z	INSULATION PLATE	1	\$ 0.24
TSTD0150007	TERMINAL:JUG SOLDER	2	\$ 0.07
TSTD0200003	SPRING PLATE:KNOB	7	\$ 0.44
VNYL1114000	VINYL BAG	1	\$ 0.60
VNYL1122800	VINYL BAG	1	\$ 0.11
VNYL2465270	VINYL BAG	0.165	\$ 1.6066
VNYL3101200	VINYL BAG	1	\$ 0.60
VNYL3324500	VINYL BAG	1	\$ 0.50
WSFC121185Z	STYROFOAM PAD (R)	1	\$ 1.20
WSFC221184A	STYROFOAM PAD (F)	1	\$ 1.36
WSL V421401Z	SPACER	1	\$ 0.26
SMR-2510 SERVICE MANUAL			

6. IC VOLTAGE CHART

TRANSISTOR

UNIT : V

REF. NO	Tx Rx	Future	AM			FM			SSB(U/L)			CW		
			E	C	B	E	C	B	E	C	B	E	C	B
Q101	RX	RF GAIN ON/OFF	1.7 2.5	6.6 7.0	2.4 2.7									
Q102	RX		3.2	7.2	0									
Q103	Tx Rx		0 0.7	0.1 3.0	0 0.7									
Q104	TX RX		2.8 0.7	8.0 5.8	2.2 1.3									
Q105	TX RX		0 2.3	0.1 6.5	0.1 3.0									
Q106	TX RX													
Q107	TX RX		0 1.6	0 6.5	0 2.3									
Q108	TX RX		0 0	0 0	0.7 0.7				0 0	0 0.5	0 0			
Q109	TX RX	SQ ON/OFF	0.6 1.1/0.6	5.0 2.8/5.0	1.0 1.3/1.1									
Q118	TX RX		0 0	7.8 8.1	0 0.46									
Q111	TX RX		0 0	0 0	0 0				0 0	0 0	0.7 0.7			
Q114	TX RX		0 0	0 0	0 0.7				0 0	0 0	0 0.7			
Q115	TX RX		0 0	0.7 0	0.1 0									
Q116	TX RX		0.3 0.4	2.8 3.0	1.6 0.7									
Q138	TX RX		8.0 8.0	0 0	7.5 7.5				8.0 8.0	0 0	7.5 7.5			
Q123		PA-PTT ON/OFF	1.2/ 1.7	6.0/ 7.0	0.8/ 1.0									
Q126	TX RX		7.4 7.4	0 0	0 0							0.3 6.5	3.4 8.0	0.9 0.6
Q117	TX RX	SQ ON/OFF	0 %	0 %	0.6 0.6%									
Q135	TX RX		0.6 0.6	3.0 3.0	0.6 0.6									
Q128	TX RX											0.5 3.5	0.5 4.8	1.1 3.6
Q125	TX RX		7.3 7.4	7.8 0	8.0 8.0									

TRANSISTOR

UNIT: V

REF. NO	Tx Rx	Future	AM			FM			SSB(U/L)			CW		
			E	C	B	E	C	B	E	C	B	E	C	B
Q127	TX RX		0.1 7.2	8.1 8.0	0 7.7									
Q136	TX RX		0 0	0 8.0	0.7 0									
Q129	TX RX		0 0	0 0	0 0	0 0	0 0	0.7 0.7	0 0	0 0	0.7 0.7			
Q134	TX RX		2.9 0	4.3 0.1	2.2 0									
Q124	TX RX		0.6 0	7.8 0	1.3 0									
Q133	TX RX		0 0	12.3 13.0	0.7 0.1									
Q132	TX RX		0 0	11.5 13.0	0.4 0									
Q131	TX RX		0 0	0 0	0 0							0 0	0 0	0.6 0.6
Q112	TX RX		0 0	0 0	0.3 0.3							0 0	0 0	0.7 0.7
Q113	TX RX		0 0	0 0	0.3 0.3							0 0	0 2.5	0.5 0
Q119	TX RX		4.6 4.8	6.5 6.5	0.1 0							4.5 5.1	6.6 7.5	0.8 0
Q121	TX RX		7.2 0	7.5 7.5	1.2 0	7.8 0	7.5 7.5	1.2 0	7.8 0	7.5 7.5	1.2 0			
Q120	TX RX		7.8 0	7.3 0	7.2 7.2	7.8 0	7.8 0	7.2 0	7.8 0	7.8 0	7.2 0			
Q106	TX RX		2.6 0	6.8 0	3.2 0				2.5 2.1	6.8 5.7	3.2 2.7	2.5 2.5	6.6 6.5	3.0 3.0
Q122	TX RX		0 0	0 0	0.2 0.2							0 0	7.2 0	0.2 0.7
Q201	RX	NB ON							0.9	7.0	1.5			
Q202	RX	NB ON							0	2.3	0.7			
Q203	RX	NB ON							1.5	6.9	2.7			
Q204	RX	NB ON							0.9	7.2	0			
Q205	RX	NB ON							0	4.2	0			
Q206	RX	NB ON							6.5	0	4.4			
Q207	RX	NB ON							0	0	0			
Q401	RX	NB ON				2.0	5.8	2.6						

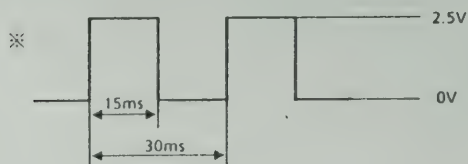
IC

UNIT : V

	IC101 NJM2902N		IC103 FDA1905	IC104 NJM4558S		IC105 AN612		IC106 TA7320P		IC107 HA17808W
		SQ			Tx		Tx		Tx	
1	0		6.8	7.6	7.2	3.0	3.0	0	6.2	13.2
2	0		13.2	4.1	4.0	3.2	3.2	0	3.3	0
3	0		12.0	2.7	2.6	3.2	3.2	0	2.7	8.0
4	8.0		4.0	4.1	4.0	0	0	0	7.6	
5	0		0.0	0	0	6.0	5.8	0	0	
6	0		2.4	3.5	3.3	7.4	7.3	0	2.7	
7	0		2.0	7.1	2.3	6.4	4.0	0	4.2	
8	0	6.8	1.5	1.3	5.7			0	4.2	
9	1.7			7.6	7.2			0	7.5	
10	1.7	6.8								
11	0									
12	0									
13	0									
14	0									

IC	UNIT: V					
	IC301	IC302	IC303	IC304	IC305	IC306
1	5.0	2.5	0	0	0	8.0
2	0	2.1	0	5.2	8.0	6.0
3	5.0	5.0	0	8.0	8.0	4.0
4	5.0	5.0	1.6	0	0	4.0
5	5.0	0	1.6	8.0	8.0	0
6	0	2.4	5.0	0	0	4.0
7	0	0	2.9	2.8	2.8	4.0
8	5.0	0	0.7	2.8	2.8	3.8~6.0
9	1.7	0		0	0	8.0
10	0			0.6	0.6	
11	0			1.3	1.3	
12	0			0.6	0.6	
13	4.0			1.3	1.3	
14	0			0	0	
15	2.5					
16	2.4					
17	2.0					
18	0					
19						
20						
21						
22						

IC	UNIT: V					
	IC307	IC311	IC312	IC314	IC316	IC501
1	8.0	13.8	13.8	5.0	5.0	※
2	7.4	0	0.6	2.0	0	※
3	1.0	8.0	5.6	0	5.0	※
4	1.0			1.2	0	※
5	0			5.0	5.0	※
6	4.2~4.9				5.0	※
7	4.2~4.9				0	※
8	3.5~4.9				0	※
9	8.0				0	※
10					5.0	※
11					5.0	※
12					0	※
13					5.0	※
14					5.0	※
15						3.5
16						1.0
17						4.2
18						4.3
19						0.7
20						0
21						0
22						0



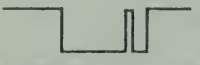
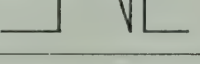
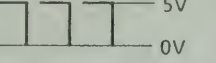
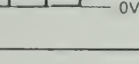
TRANSISTOR

UNIT : V

	E (S)	C (D)	B (G)	REMARKS
Q301	2.5	8.0	0.9	
Q302	2.5	7.8	4.0	
Q303	4.0	8.0	4.6	
Q304	2.9	8.0	3.5	
Q305	0.2	7.7	0	
Q306	0.3	3.9	0	
Q307	0.2	3.8	0	
Q308	4.0	5.3	4.5	
Q311	0 0	0 0	0 0.5	Rx / Tx Meter RF Tx other
Q312	0 0	0 0	0 0.5	Tx Meter MOD Rx / Tx other
Q313	0 0	0 0	0 0.5	Tx Meter CAL Rx / Tx other
Q314	0 0	0 0	0 0.5	Tx Meter SWR Rx / Tx other
Q315	0 0	7.3 0.7	0 0.7	Rx Tx
Q316	0	0.5	0	
Q317	5.0	0	5.0	

IC

UNIT : V

IC315					
Pin No.	Voltage		Pin No.	Voltage	
1	5	LCD Change 	17	5	
			18	0	
			19	5	
2	0	LCD Change 	20	0	
			21	0	
3	5	LCD Change 	22	5	(F.Knob)
			23	0	(F.Knob)
			24	5	
4	0	LCD Change 	25	5	(USB)
			26	5	(LSB)
			27	5	(CW)
5	5		28	0	5V pulse (Freq. & TxRx Change)
6	5	0 (Tx Meter RF)	29	0	5V pulse (Freq. & TxRx Change)
7	5	0 (Tx Meter MOD)	30	0	5V pulse (Freq. & TxRx Change)
8	5	0 (Tx Meter CAL)	31	0	5V pulse (Freq. & TxRx Change)
9	5	0 (Tx Meter SWR)	32	5	
10	5	Up/Down/Span/Scan/Meter/Band SW 	33		
11	5		34	0	
12	5		35	0	
13	5		36	5	
14	0	Band/Span/Down ON			
15	0	Meter/Scan/Up ON			
16	0	Band/Span/Down ON			

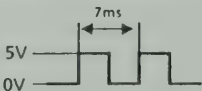
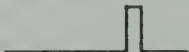
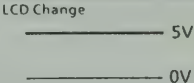
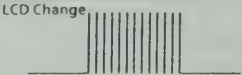
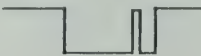
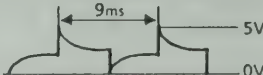
IC

UNIT: V

IC315											
Pin No.	0* 0.0KHz	0*	1*	2*	3*	4*	5*	6*	7*	8*	9*
37		0	0	0	0	5	5	5	5	5	5
38		0	5	5	5	0	0	0	5	5	5
39		5	0	5	5	0	5	5	0	5	5
40		5	0	0	5	5	0	5	0	0	5
41		0	0	0	5	0	0	5	5	0	0
42		0	0	0	0	0	5	0	5	0	5
Pin No.	Voltage					Pin No.	Voltage				
43	0					55	0 (SQ Open)	5 (SQ Close)			
44	0					56	5				
45	5	0	(Push Mic up SW)				57	0			
46	5	0	(Push Mic Down SW)				58	5			
47	5	0	(PA SW OFF PTT ON Tx)				59	5			
48	5	0	(F.LOCK SW ON)				60	5			
49	0	5	(Power ON)				61	0	LCD Change		
50	5										
51	2.5	4 V _{p-p} (2.00MHz)									
52	2.5	4 V _{p-p} (2.00MHz)				62	5 (Tx)	0 (Rx)			
53	0					63	(Tx) 5 (CW)	0 (Rx)	5 (SQ)	0	
54	0					64	5				

IC

UNIT: V

IC502										
Pin No.		Voltage			Pin No.		Voltage			
1					11		0			
2					12		0			
3		2.4			13		5			
4		2.6			14					
5		1.8			15					
6		5.0			18					
7		0			19~32					
8		5				33		5.0		
9		0				34~51				
10		5				52				

7. FREQUENCY CHART

FREQUENCIES OF LOCAL OSCILLATORS

$$F_v = F + 10.6950 \text{ (MHz)} + \alpha$$

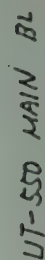
$$\left(\begin{array}{l} F : \text{TRANSMITTING AND RECEIVING FREQUENCY} \\ F_v : \text{VCO OUTPUT FREQUENCY} \\ \alpha \left\{ \begin{array}{ll} 0 & \text{(CW / FM / AM)} \\ + 2.5 \text{ kHz} & \text{(USB)} \\ - 2.5 \text{ kHz} & \text{(LSB)} \end{array} \right. \end{array} \right)$$

CHANNEL AND FREQUENCY RANGE

BAND	FREQUENCY RANGE
A	28.0000 ~ 28.4999 MHz
B	28.5000 ~ 28.9999 MHz
C	29.0000 ~ 29.4999 MHz
D	29.5000 ~ 29.6999 MHz

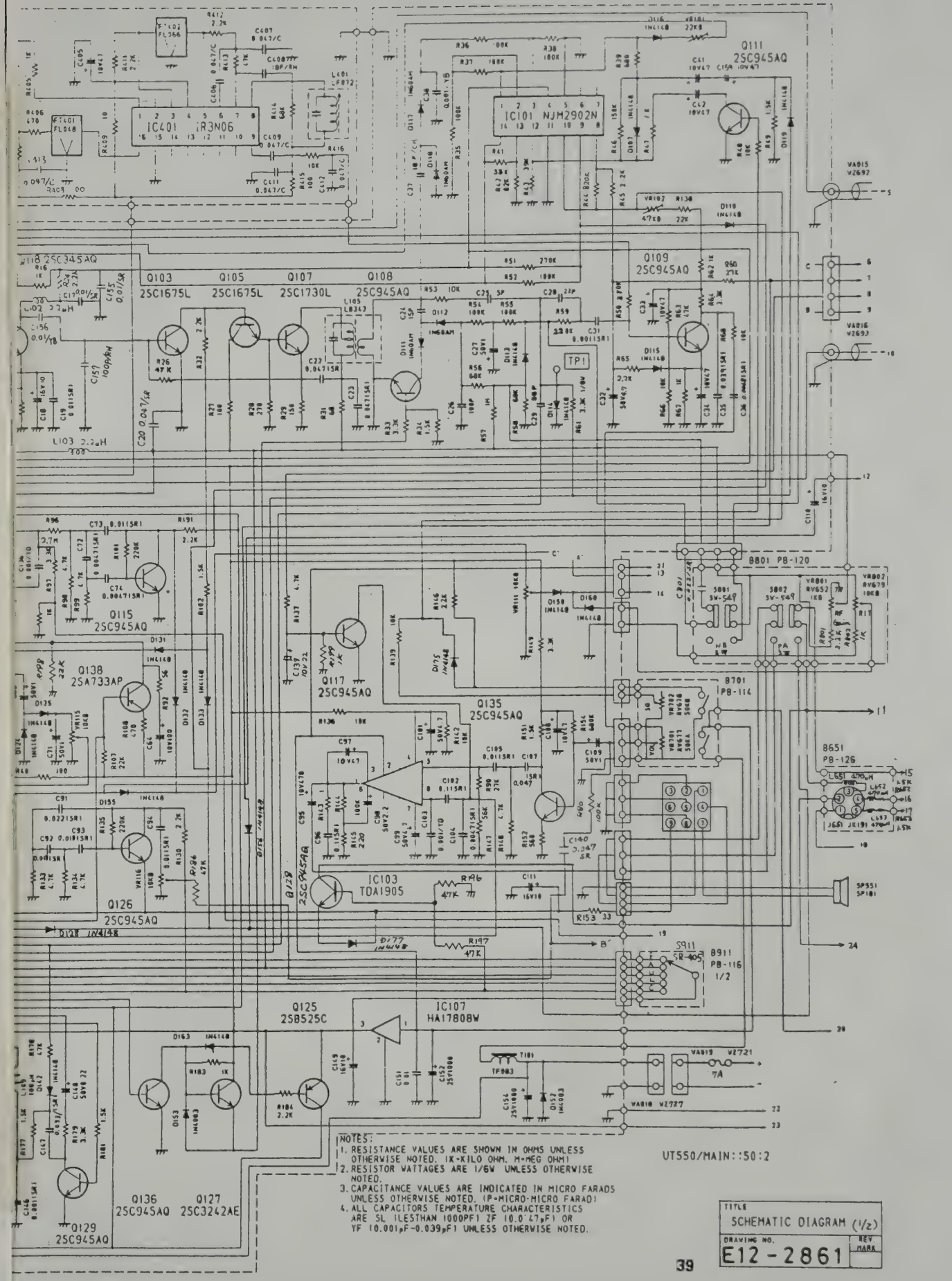
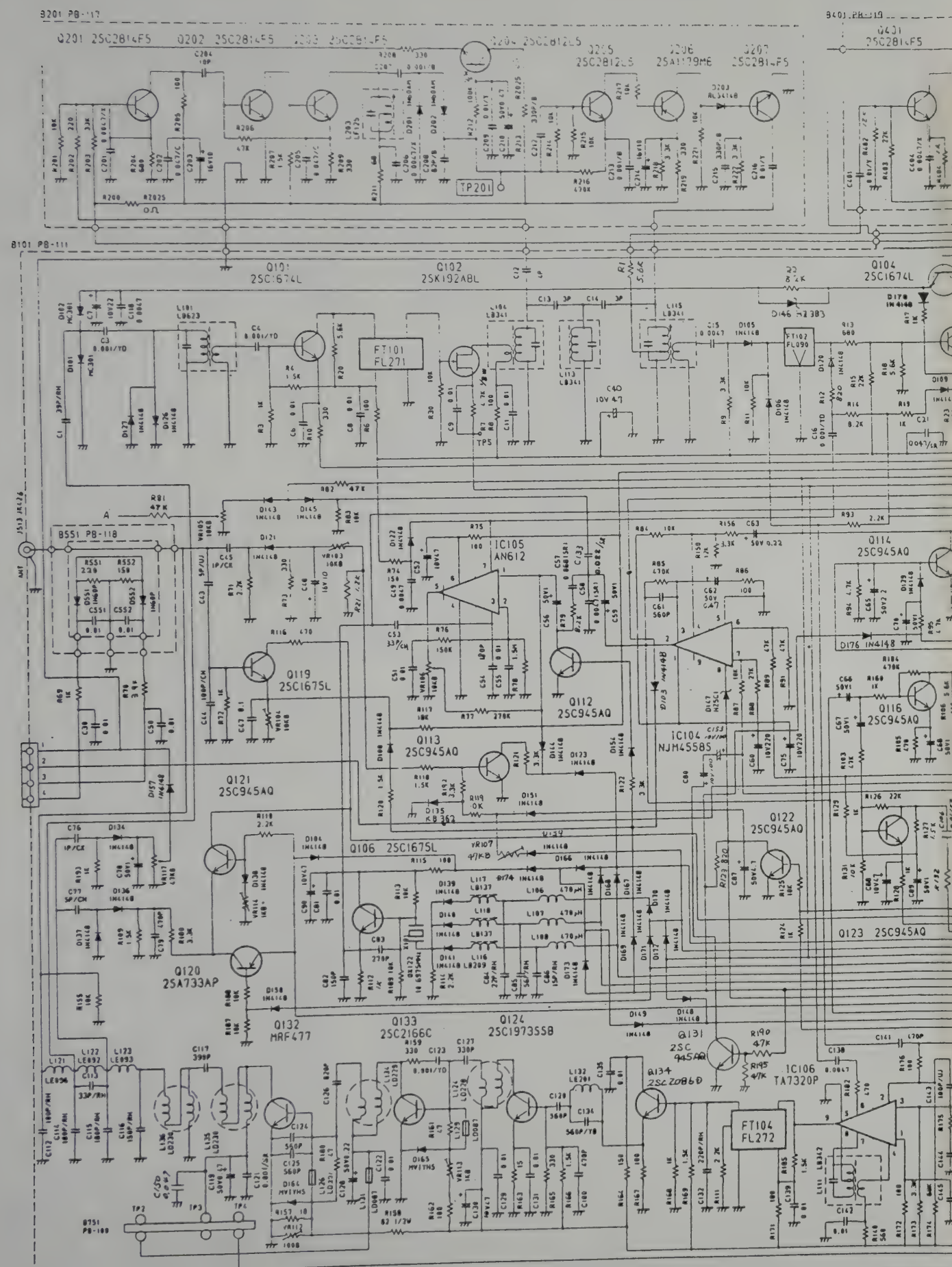
8. TECHNICAL DRAWINGS

UT-550 MAIN BL DIAGRAM

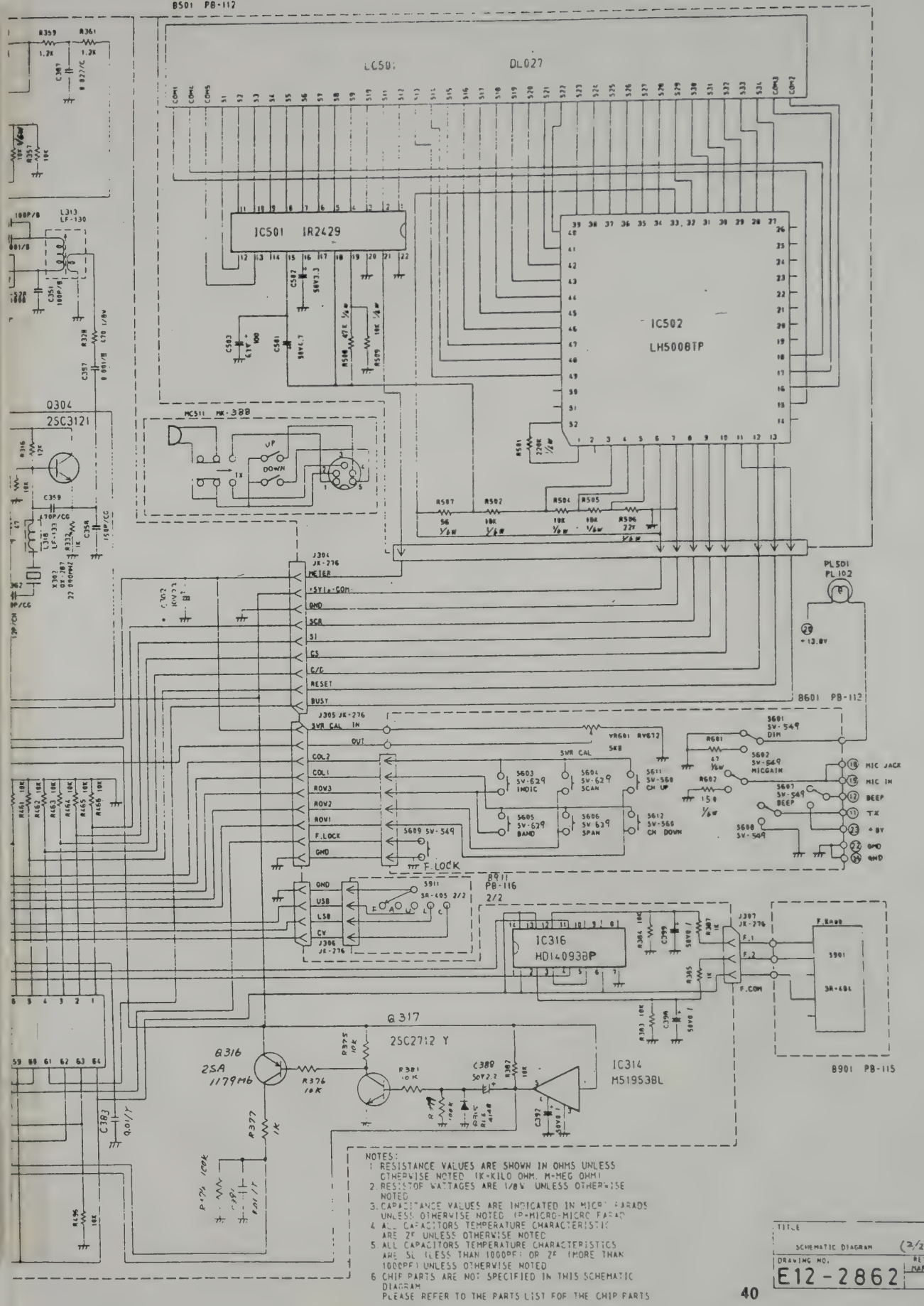
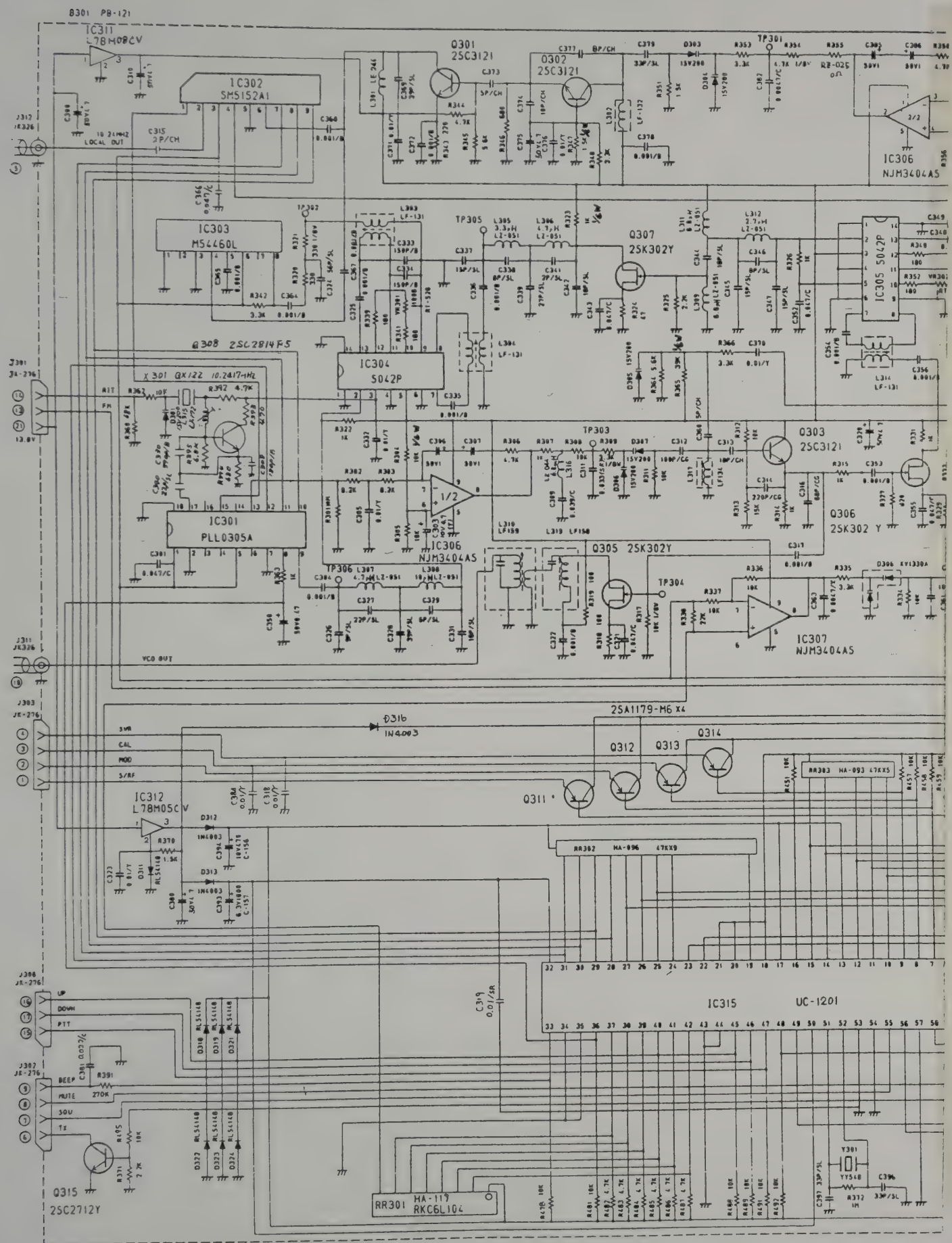


UT-550 MAIN BL DIAGRAM

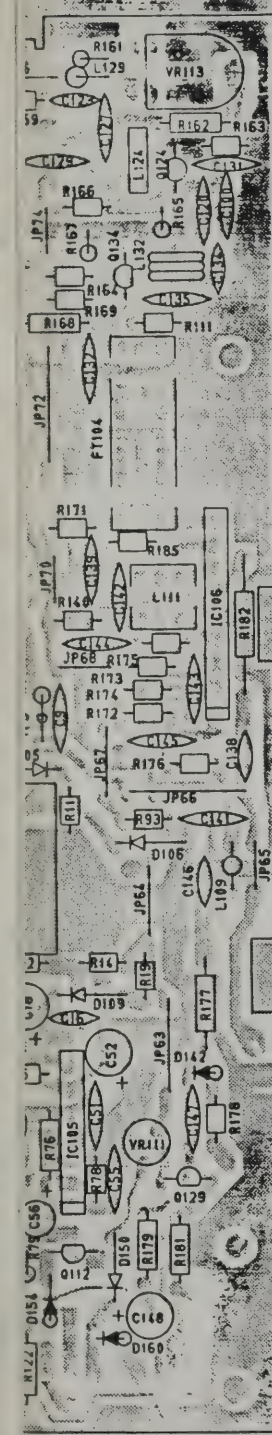
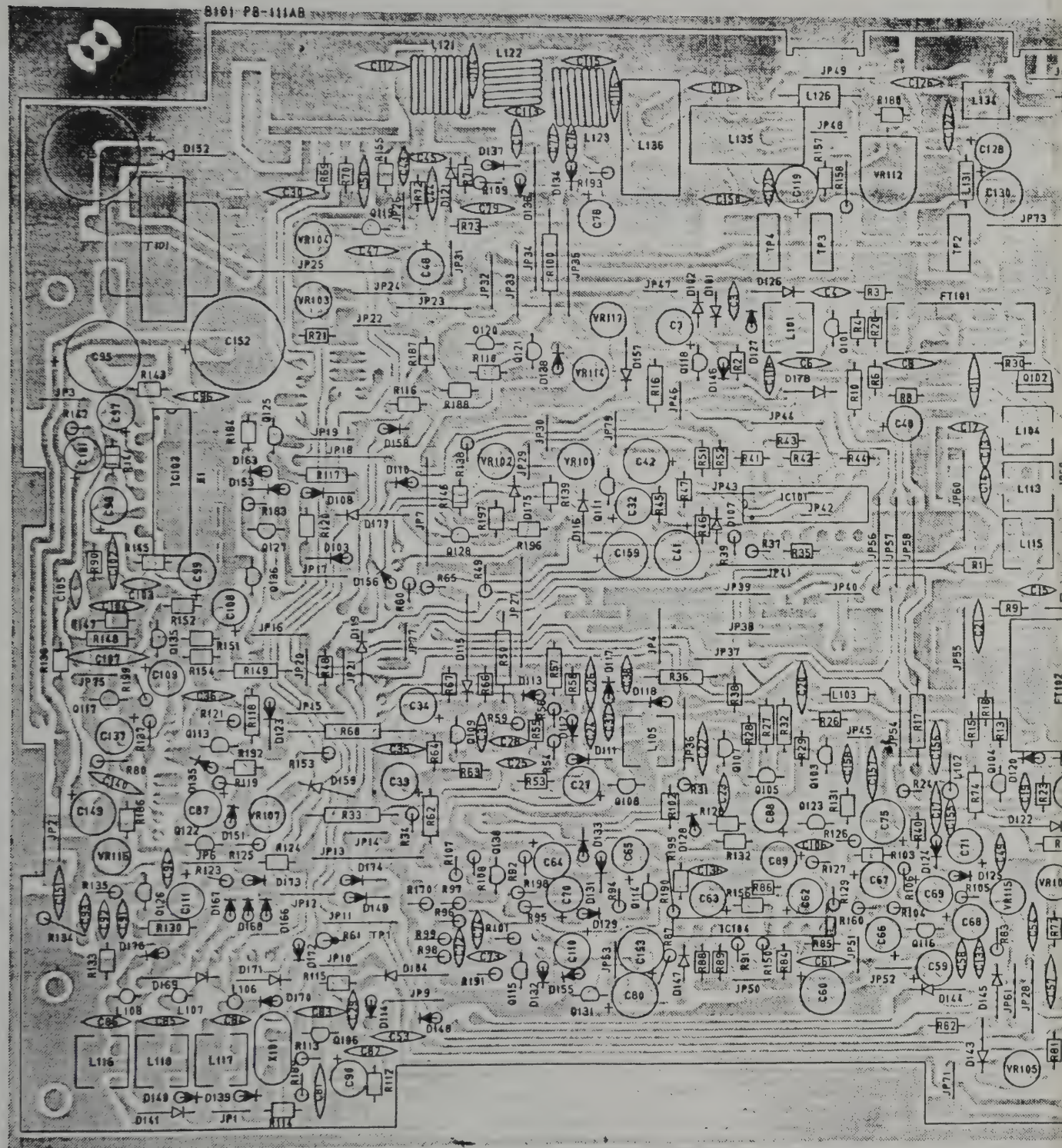




NOTES:
 1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. 1K=KILO OHM, M=MEG OHM
 2. RESISTOR VATTAGES ARE 1/6W UNLESS OTHERWISE NOTED
 3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P=PICTO-MICRO FARAD)
 4. ALL CAPACITORS TEMPERATURE CHARACTERISTICS ARE SL (LESTHAN 1000PF) 2F (10.0-47PF) OR YF (10.001PF-0.039PF) UNLESS OTHERWISE NOTED.

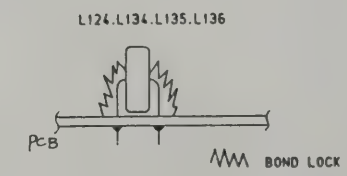
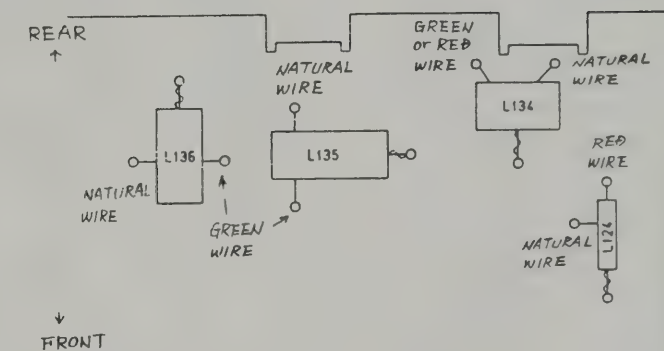


NOTES:
 1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. 1K=KILO OHM, M=MEG OHM.
 2. RESISTOR WATTAGES ARE 1/8W UNLESS OTHERWISE NOTED.
 3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P=PICTO-FARAD)
 4. ALL CAPACITORS TEMPERATURE CHARACTERISTICS ARE 2% UNLESS OTHERWISE NOTED.
 5. ALL CAPACITORS TEMPERATURE CHARACTERISTICS ARE 5% UNLESS OTHERWISE NOTED.
 6. CHIP PARTS ARE NOT SPECIFIED IN THIS SCHEMATIC DIAGRAM.
 PLEASE REFER TO THE PARTS LIST FOR THE CHIP PARTS.

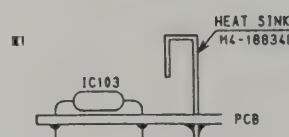


NOTE

① L124, L134, L135, L136



②



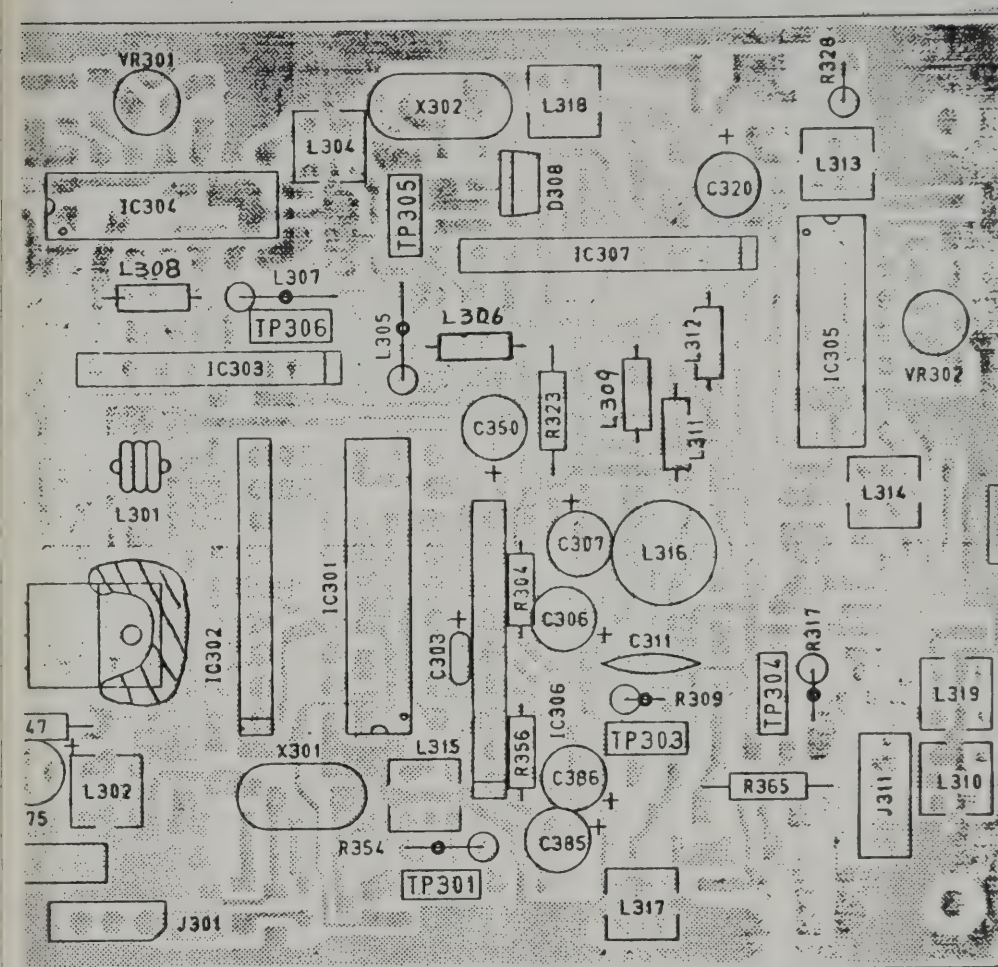
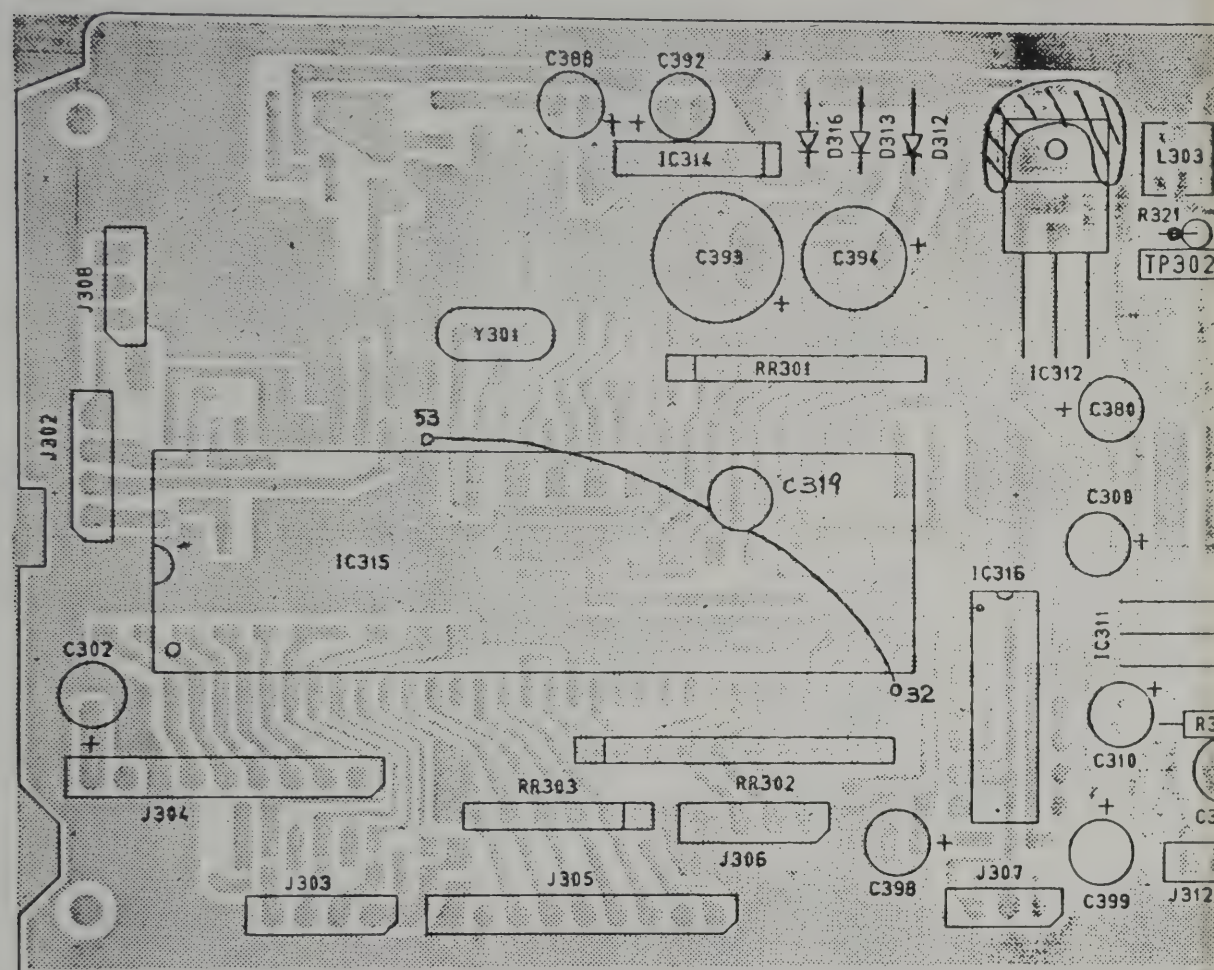
C1	39P/RH	C72	0.0047(SR)	C140	0.047(SR)	R1	5.6K	R70	3.9K	R136	10K	D101	MC301	D172	IN4148	JP1	500	JP50	500
C3	0.001/YD	C73	0.01(SR)	C141	470P	R2	8.2K	R71	2.2K	R137	4.7K	D102	MC301	D173	IN4148	JP2	1000	JP51	500
C4	0.001/YD	C74	0.0047(SR)	C142	0.01	R3	1K	R72	1K	R138	22K	D103	IN4148	D174	IN4148	JP3	600	JP52	500
C6	0.01	C75	10V220 C-095	C143	100P/UJ	R4	1.5K	R73	330	R139	10K	D104	IN4148	D175	IN4148	JP4	7.500	JP53	500
C7	10V22	C76	1P/CK	C144	330P/UJ	R6	100	R74	150	R140	560	D105	IN4148	D176	IN4148	JP6	7.500	JP54	1000
C8	0.01	C77	5P/CH	C145	0.01	R7	4.7K 1/8W	R75	100	R142	10K	D106	IN4148	D177	IN4148	JP7	1500	JP55	12.500
C9	0.01	C78	50V1	C146	0.001(SR)	R8	100	R76	150K	R143	1	D107	IN4148	D178	IN4148			JP56	12.500
C11	0.01	C79	470P	C147	0.033(SR)	R9	3.3K	R77	270K	R144	100K	D108	IN4148			JP9	7.500	JP57	12.500
C12	4P	C80	10V100 C-156	C148	50V0.22	R10	330	R78	1.5M	R145	220	D109	IN4148			JP10	500	JP58	12.500
C13	3P	C81	0.01	C149	16V10 C-095	R11	10K	R79	8.2K	R146	22K	D110	IN4148			JP11	1000	JP60	12.500
C14	3P	C82	150P	C150	0.047/ZF	R12	820	R80	100K	R147	56K	D111	IN60AM	L101	LB623	JP12	7.500	JP61	7.500
C15	0.0047	C83	270P	C151	0.01	R13	680	R81	47K	R148	4.7K	D112	IN60AM	L102	LZ051 2.2, H	JP13	1500	JP63	1000
C16	0.001/YD	C84	22P/RH	C152	25V1000 C-095	R14	8.2K	R82	47K	R149	3.3K	D113	IN4148	L103	LZ051 2.2, H	JP14	500	JP64	7.500
C17	0.01(SR)	C85	56P/RH	C153	10V100 C-156	R15	22K	R83	10K	R150	12K	D114	IN4148	L104	LB341	JP15	7.500	JP65	500
C18	16V10	C86	15P/RH	C154	25V1000 C-095	R16	1K	R84	10K	R151	1.5K	D115	IN4148	L105	LB342	JP16	7.500	JP66	12.500
C19	0.01(SR)	C87	50V4.7	C155	0.01(SR)	R17	1K	R85	470K	R152	560	D116	IN4148	L106	LZ016 470, H	JP17	7.500	JP67	7.500
C20	0.047(SR)	C88	10V47	C156	0.01/YB	R18	5.6K	R86	100	R153	33	D117	IN60AM	L107	LZ016 470, H	JP18	12.500	JP68	500
C21	0.047(SR)	C89	50V1	C157	100P/RH	R19	1K	R87	10K	R154	680K	D118	IN60AM	L108	LZ016 470, H	JP19	7.500	JP69	7.500
C22	0.047(SR)	C90	10V47	C159	10V47 C-095	R20	5.6K	R88	27K	R155	10K	D119	IN4148	L109	LZ016 100, H	JP20	500	JP70	500
C23	0.047(SR)	C91	0.022(SR)			R21	1.2K	R89	47K	R156	3.3K	D120	IN4148	L111	LB342	JP21	500	JP71	500
C24	15P	C92	0.018(SR)			R23	680	R90	27K	R157	10	D121	IN4148	L113	LB341	JP22	500	JP72	12.500
C25	5P	C93	0.018(SR)			R24	2.2K	R91	47K	R158	82 1/2W	D122	IN4148	L115	LB341	JP23	1000	JP73	12.500
C26	100P	C94	0.01(SR)			R26	47K	R92	56	R159	330	D123	IN4148	L116	LB209	JP24	1000	JP74	500
C27	50V1	C95	10V470 C-095			R27	100	R93	2.2K	R160	1K	D124	IN4148	L117	LB137	JP25	2000	JP75	500
C28	22P	C96	0.1(SR)			R28	270	R94	4.7K	R161	47	D125	IN4148	L118	LB137	JP26	500	JP76	1000
C29	56P	C97	10V47 C-095			R29	150	R95	4.7K	R162	100	D126	IN4148	L121	LE096	JP27	1500	JP77	500
C30	0.01	C98	50V2.2			R30	10K	R96	2.7M	R163	15	D127	IN4148	L122	LE092	JP28	1000	JP79	500
C31	0.001(SR)	C99	50V4.7 C-095			R31	68	R97	3.3K	R164	150	D128	IN4148	L123	LE093	JP29	500		
C32	50V4.7	C100	470P			R32	2.2K	R98	4.7K	R165	330	D129	IN4148	L124	LD228	JP30	500		
C33	10V47	C101	50V4.7 C-095			R33	3.3K	R99	4.7K	R166	1.5K	D131	IN4148	L126	LD221	JP31	500		
C34	10V47	C102	0.1(SR)	Q101	25C1674L	R34	1.5K	R100	3.3K	R167	100	D132	IN4148	L129	LD087	JP32	7.500		
C35	0.039(SR)	C103	0.001/YD	Q102	25K192ABL	R35	100K	R101	220K	R168	1K	D133	IN4148	L131	LD087	JP33	1000	FT101	FL271
C36	0.0068(SR)	C104	0.0047(SR)	Q103	25C1675L	R36	100K	R102	1.5K	R169	1.5K	D134	IN4148	L132	LE201	JP34	2000	FT102	FL090
C37	18 P/CH	C105	0.01(SR)	Q104	25C1674L	R37	100K	R103	47K	R170	1K	D135	KB362	L134	LD229	JP35	1500	FT104	FL272
C38	0.001/YB	C106	0.01(SR)	Q105	25C1675L	R38	100K	R104	470K	R171	100	D136	IN4148	L135	LD230	JP36	1000		
C40	10V47	C107	0.047(SR)	Q106	25C1675L	R39	680	R105	470	R172	100	D137	IN4148	L136	LD230	JP37	1000		
C41	10V47 C-095	C108	10V47	Q107	25C1730L	R40	100	R106	5.6K	R173	3.3K	D138	IN4148			JP38	7.500		
C42	10V47	C109	50V1	Q108	25C945AQ	R41	33K	R107	22K	R174	68K	D139	IN4148			JP39	7.500		
C43	5P/UJ	C110	16V10 C-095	Q109	25C945AQ	R42	82K	R108	470	R175	33	D140	IN4148			JP40	500	X101	QX122
C44	100P/CH	C111	16V10	Q110	25C945AQ	R43	39K	R109	1.5K	R176	100	D141	IN4148			JP41	2000		
C45	1P/CK	C112	100P/RH	Q111	25C945AQ	R44	820K	R110	2.2K	R177	1.5K	D142	IN4148	IC101	NJM2902N	JP42	500		
C47	0.1/ZF	C113	33P/RH	Q112	25C945AQ	R45	2.2K	R111	2.2K	R178	47K	D143	IN4148	IC103	TDA1905	JP43	7.500	T101	TF083
C48	16V10	C114	180P/RH	Q113	25C945AQ	R46	150K	R112	1K	R179	3.3K	D144	IN4148	IC104	NJM4558S	JP44	1000		
C49	0.0047	C115	180P/RH	Q114	25C945AQ	R47	1K	R113	10K	R180	47	D145	IN4148	IC105	AN612	JP45	3.500		
C50	0.01	C116	150P/RH	Q115	25C945AQ	R48	10K	R114	2.2K	R181	1.5K	D146	HZ3B3	IC106	TA7320P	JP46	500	TP2	TP044
C51	0.01	C117	390P	Q116	25C945AQ	R49	1.5K	R115	100	R182	470	D147	HZ5C1			JP47	500	TP3	TP044
C52	10V47	C118	0.0047	Q117	25C945AQ	R50	270K	R116	470	R183	1K	D148	IN4148			JP48	500	TP4	TP044
C53	33P/CH	C119	50V0.47 C-095	Q118	25C1675L	R51	270K	R117	18K	R184	2.2K	D149	IN4148			JP49	1000		
C54	180P	C120	560P	Q119	25C1675L	R52	100K	R118	1.5K	R185	1.5K	D150	IN4148						
C55	0.01	C121	0.001(SR)	Q120	25A733AP	R53	10K	R119	10K	R186	47K	D151	IN4148	VR101	RT528 22KB				
C56	50V1 C-095	C122	0.01	Q121	25C945AQ	R54	100K	R120	1.5K	R187	10K	D152	IN4003	VR102	RT528 47KB				
C57	0.068(SR)	C123	0.001/YD	Q122	25C945AQ	R55	100K	R121	3.3K	R188	10K	D153	IN4003	VR103	RT528 10KB				
C58	0.0047(SR)	C126	820P	Q123	25C19735SB	R56	68K	R122	3.3K	R189	10K	D154	IN4148	VR104	RT528 10KB				
C59	50V1	C127	330P	Q124	25C19735SB	R57	1M	R123	820	R190	47K	D155	IN4148	VR105	RT528 10KB				
C60	10V220 C-095	C128	50V0.22 C-095	Q125	25C945AQ	R58	68K	R124	1K	R191	2.2K	D156	IN4148	VR106	RT528 10KB				
C61	560P	C129	0.01	Q126	25C945AQ	R59	220K	R125	10K	R192	3.3K	D157	IN4148	VR107	RT528 47KB				
C62	50V0.47	C130	10V47 C-095	Q127	25C3242AE	R60	27K	R126	22K	R193	1K	D158	IN4148	VR111	RT528 10KB				
C63	50V0.22	C131	0.01	Q128	25C945AQ	R61	3.3K 1/8W	R127	1.5K	R195	47K	D159	IN4148	VR112	RT182 100B				
C64	10V100 C-156	C132	220P/RH	Q129	25C945AQ	R62	1K	R128	1K	R196	47K	D160	IN4148	VR113	RT182 1KB				
C65	50V2.2	C133	0.022(SR)	Q130	25C945AQ	R63	47K	R129	1K	R197	47K	D163	IN4148	VR114	RT528 1KB				
C66	50V1	C134	560P/YB	Q131	25C2086D	R64	3.3K	R130	2.2K	R198	22K	D166	IN4148	VR115	RT528 10KB				
C67	50V1	C135	0.01	Q132	25C945AQ	R65	2.2K	R131	10K	R199	1K	D167	IN4148	VR116	RT528 10KB				
C68	50V1 C-095	C136	0.001/YD	Q133	25C945AQ	R66	10K	R132	470K			D168	IN4148	VR117	RT528 47KB				
C69	50V1 C-095	C137	10V22 C-095	Q134	25C945AQ	R67	1K	R133	4.7K			D169	IN4148						
C70	50V1 C-095	C138	0.0047	Q135	25A733AP	R68	10K	R134	4.7K			D170	IN4148						
C71	50V1	C139	0.01	Q136		R69	1K	R135	220K			D171	IN4148						

NOTES:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K-KILO OHM, M-MEG OHM)
2. RESISTOR WATTAGES ARE 1/8W UNLESS OTHERWISE NOTED.
3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P-MICRO-MICRO FARAD)
4. ALL CAPACITORS TEMPERATURE CHARACTERISTICS ARE SL (LESS THAN 1000PF) OR YF (MORE THAN 1000PF) UNLESS OTHERWISE NOTED.

TITLE MAIN PCB
PARTS ASSEMBLY TOP VIEW

E23-7563 2/ REV. MARK



C300	50V4.7
C302	10V22
C303	10V4.7IT1
C306	50V1
C307	50V1
C310	50V4.7
C311	0.033(SR1
C320	50V4.7
C350	50V0.47
C375	50V4.7
C380	50V4.7
C385	50V1
C386	50V1
C388	50V2.2
C392	50V0.1
C393	6.3V1000 C-157
C394	10V470 C-156
C398	50V0.1
C399	50V0.1
C319	0.01 (SR)

[illegible]

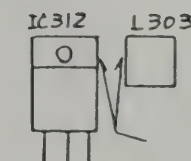
L301	LE246
L302	LF132
L303	LF131
L304	LF131
L305	LZ051 3.3 μ
L306	LZ051 4.7 μ
L307	LZ051 4.7 μ
L308	LZ051 10 μ
L309	LZ051 6.8 μ
L310	LF159
L311	LZ051 6.8 μ
L312	LZ051 2.7 μ
L313	LF130
L314	LF131
L315	LF172
L316	LZ044 6.8m
L317	LF134
L318	LF133
L319	LF158

IC301	PLL0305A
IC302	SM5152A1
IC303	M54460L
IC304	S042P
IC305	S042P
IC306	NJM3404AS
IC307	NJM3404AS
IC311	L78M08CV
IC312	L78M05CV
IC314	M51953BL
IC315	UC12D1
IC316	HD14093BP

301	JK276	3P
302	JK276	4P
303	JK276	4P
304	JK276	9P
305	JK276	9P
306	JK276	4P
307	JK276	3P
308	JK276	3P
311	JK326	
312	JK326	

R301	HA117 RKC6L104
R302	HA096 47Kx5
R303	HA093 47Kx5

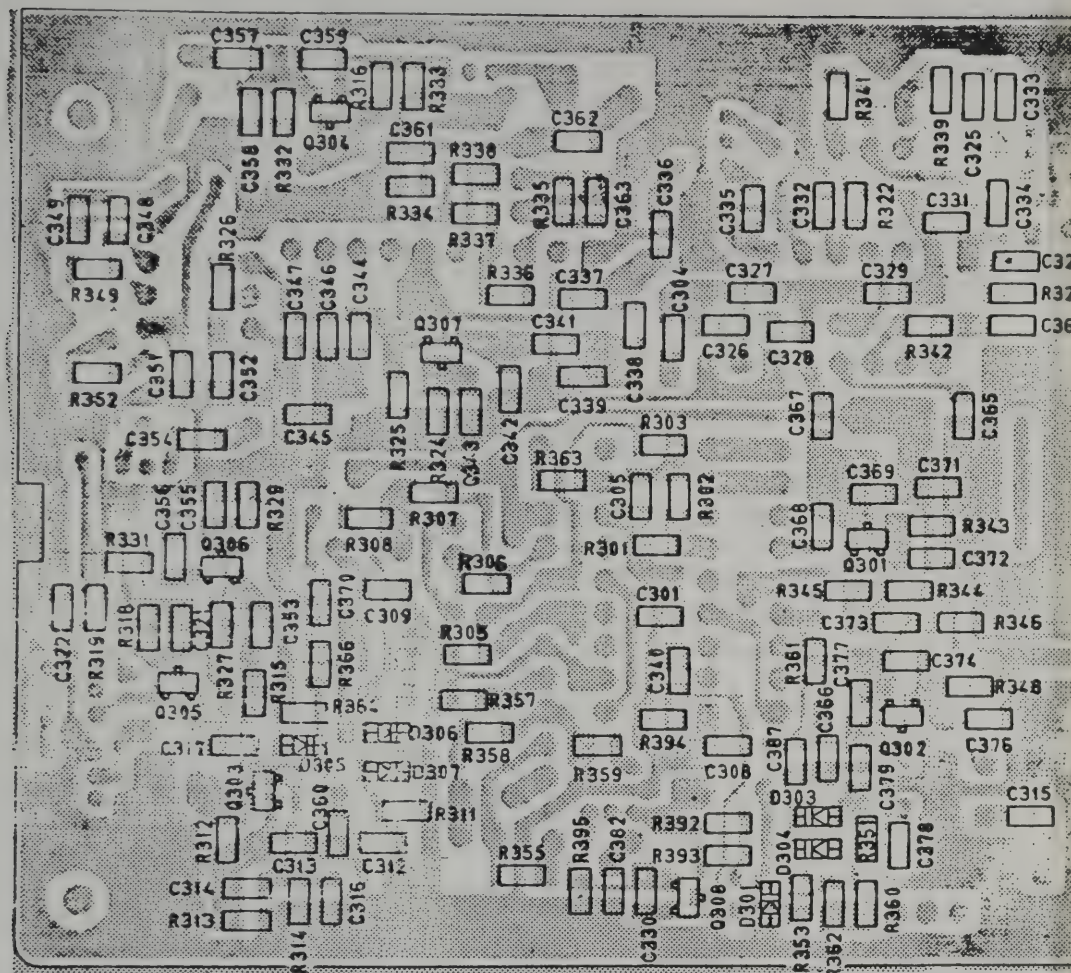
R301	RT528 100B
R302	RT528 100B
301	YY548



NOTES:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K-KILO OHM. M-MEG OHM)
2. RESISTOR WATTAGES ARE 1/6W UNLESS OTHERWISE NOTED.
3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTE. (P-MICRO - MICRO FARADS)

B301 PB121BA(BOTTOM VIEW)



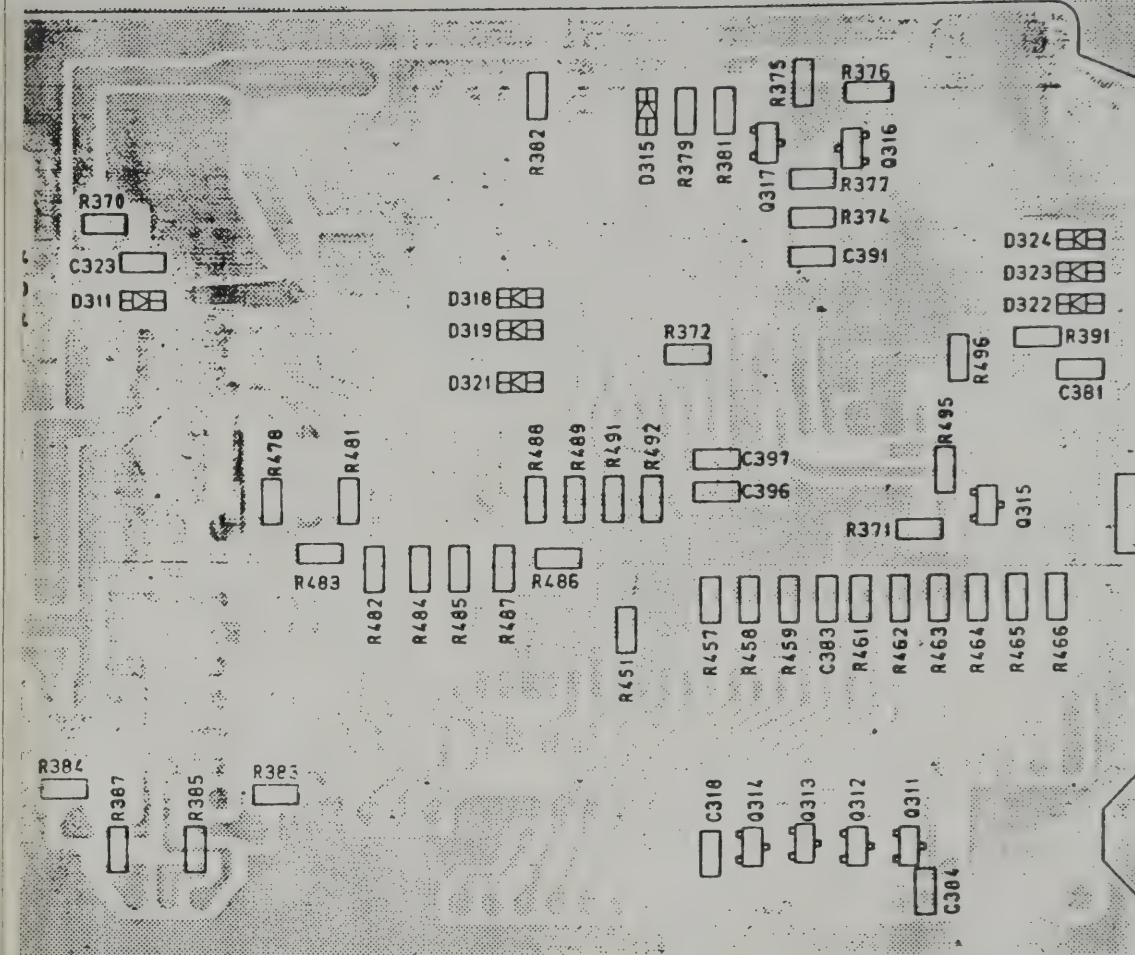
R301	10M (3216)
R302	8.2K
R303	8.2K
R305	10K
R306	4.7K
R307	1K
R308	10K
R311	10K
R312	10K
R313	15K
R314	1K
R315	1K
R316	12K
R318	100
R319	100
R320	330
R322	1K
R324	47
R325	2.2K
R326	1K
R327	820
R329	47
R331	1K
R332	1K
R333	10K
R334	10K
R335	3.3K
R336	10K

R337	10K
R338	22K
R339	180
R341	180
R342	3.3K
R343	220
R344	4.7K
R345	5.6K
R346	680
R348	3.3K
R349	180
R351	1.5K
R352	180
R353	3.3K
R355	R2025
R357	10K
R358	4.7K
R359	1.2K
R360	68K
R361	1.2K
R362	10K
R363	1K
R364	5.6K
R366	3.3K
R370	1.5K
R371	2.2K
R372	1M
R374	100K

R375	10K
R376	10K
R377	1K
R379	100K
R381	10K
R382	10K
R383	10K
R384	10K
R385	1K
R387	1K
R391	270K
R392	4.7K
R393	470
R394	680
R395	6.8K
R451	10K
R457	10K
R458	10K
R459	10K
R461	10K
R462	10K
R463	10K
R464	10K
R465	10K
R466	10K
R478	10K

R481	10K
R482	4.7K
R483	4.7K
R484	4.7K
R485	4.7K
R486	4.7K
R487	4.7K
R488	10K
R489	10K
R491	10K
R492	10K
R495	10K
R496	10K

C301	0.047/C
C304	0.001/B
C305	0.01/Y
C308	180P/B
C309	0.039/C
C312	100P/CG
C313	18P/CH
C314	220P/CG
C315	2P/CH
C316	68P/CG
C317	0.001/B
C318	0.01/Y
C321	0.047/C
C322	0.001/B
C323	0.01/Y
C324	56P/SL
C325	0.001/B
C326	9P/SL
C327	22P/SL
C328	39P/SL
C329	6P/SL
C330	390P/B
C331	18P/SL
C332	0.01/Y
C333	150P/B
C334	150P/B
C335	0.001/B
C336	0.001/D



C337	15P/SL
C338	8P/SL
C339	27P/SL
C340	22P/SL
C341	2P/SL
C342	18P/SL
C343	0.047/C
C344	18P/SL
C345	15P/SL
C346	8P/SL
C347	15P/SL
C348	0.001/B
C349	100P/B
C351	100P/B
C352	0.047/C
C353	0.001/B
C354	0.001/B
C355	0.047/C
C356	0.001/B
C357	0.001/B
C358	150P/CG
C359	470P/CG
C360	5P/CH
C361	12P/CH
C362	100P/CG
C363	0.0047/X
C364	0.001/B
C365	0.001/D

C366	0.047/C
C367	0.001/B
C368	0.001/B
C369	39P/SL
C370	0.01/Y
C371	0.01/Y
C372	0.001/B
C373	5P/CH
C374	10P/CH
C376	0.01/Y
C377	8P/CH
C378	0.001/B
C379	33P/SL
C381	0.022/C
C382	0.0047/C
C383	0.01/Y
C384	0.01/Y
C387	0.022/C
C391	0.01/Y
C396	33P/SL
C397	33P/SL

D301	1SV200
D303	1SV200
D304	1SV200
D305	1SV200
D306	1SV200
D307	1SV200
D311	RLS4148
D315	RLS4148
D318	RLS4148
D319	RLS4148
D321	RLS4148
D322	RLS4148
D323	RLS4148
D324	RLS4148

X-C-005

- NOTES:
1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KILO OHM, M=MEG OHM)
 2. RESISTOR WATTAGES ARE 1/8W UNLESS OTHERWISE NOTED.
 3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTE (P= MICRO-MICRO FARAD)

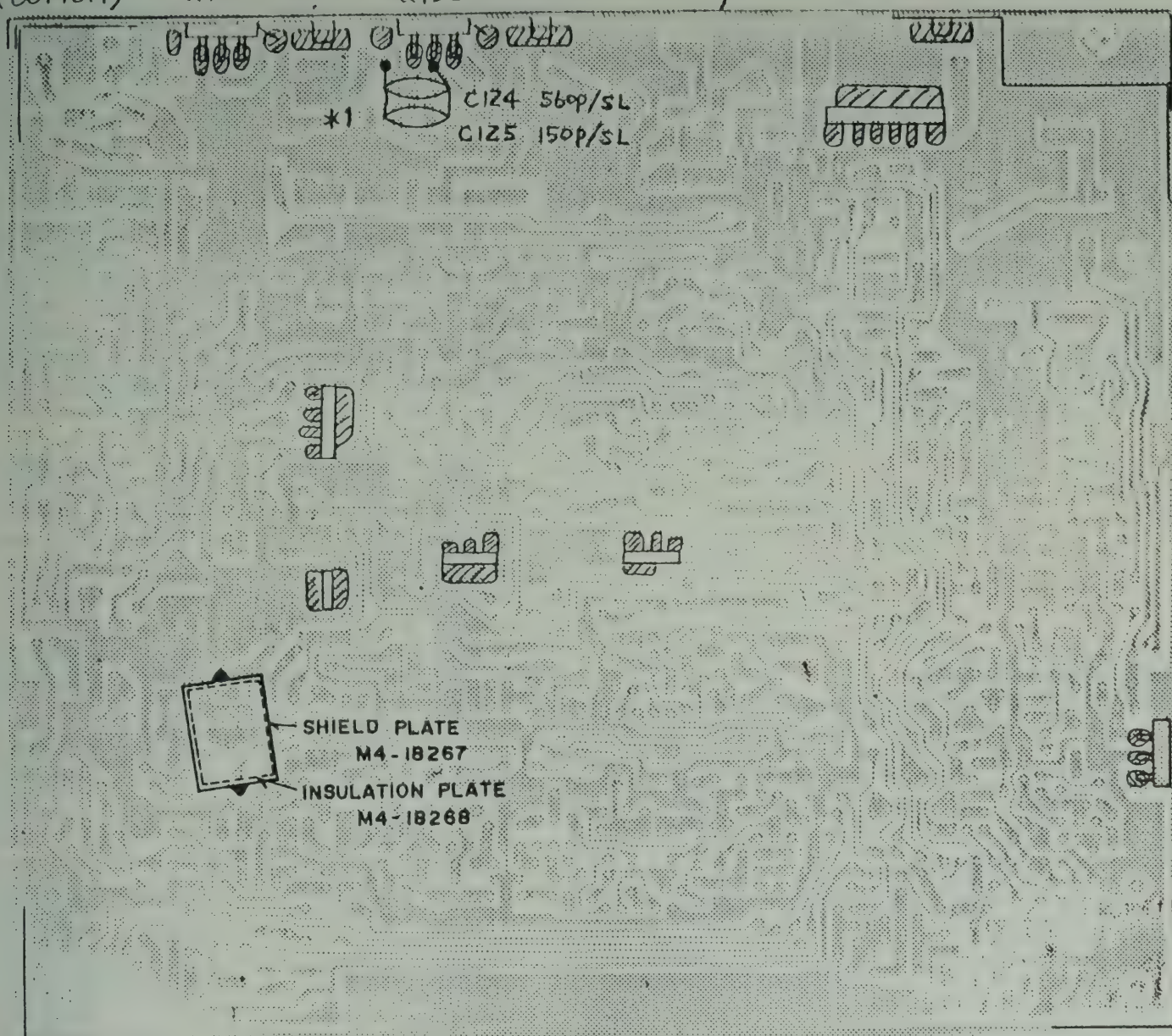
TITLE	PLL PCB
PARTS ASSEMBLY	BOTTOM VIEW
E23-7565	REV. MARK

B101
PB-III
(BOTTOM)

D165
Q133

D164
Q132

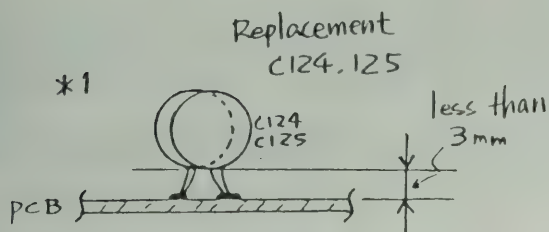
CHASSIS



 : SOLDERING

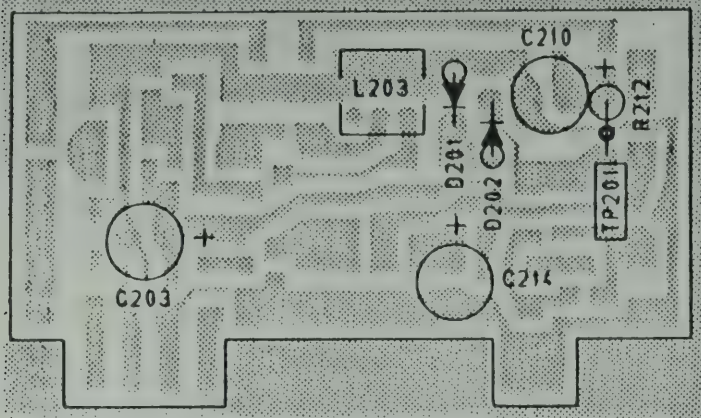
NOTE:

1. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS
UNLESS OTHERWISE NOTED. (P=MICRO-MICRO FARAD)



TITLE MAIN PCB	
PARTS ASS'Y (BOTTOM)	
DRAWING NO.	REV. MARK
E24-7566	

B201 PB-117AA (TOP VIEW)



C203	16V10
C210	50V0.47 C-095
C214	16V10

D201	1N60AM
D202	1N60AM

R212	100K

L203	LF125

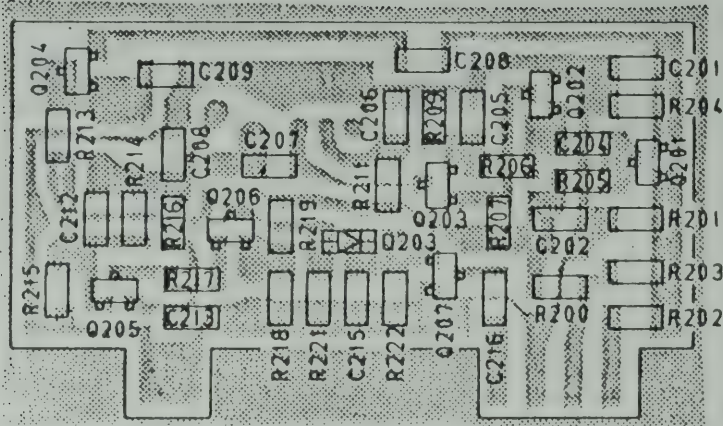
R200	R2025
R201	10K
R202	220
R203	33K
R204	680
R205	100
R206	47K
R207	1.5K
R208	330
R209	330
R211	68
R213	R2025
R214	10K
R215	10K
R216	470K
R217	10K
R218	3.3K
R219	330
R221	10K
R222	3.3K

C201	0.0047/X
C202	0.047/C
C204	10P/SL
C205	0.047/C
C206	0.0047/X
C207	0.001/B
C208	82P/B
C209	0.01/Y
C212	330P/B
C213	0.001/B
C215	330P/B
C216	0.01/Y

Q201	2SC2814F5
Q202	2SC2814F5
Q203	2SC2814F5
Q204	2SC2812L5
Q205	2SC2812L5
Q206	2SA1179M6
Q207	2SC2814F5

D203	RL54148

B201 (BOTTOM VIEW)

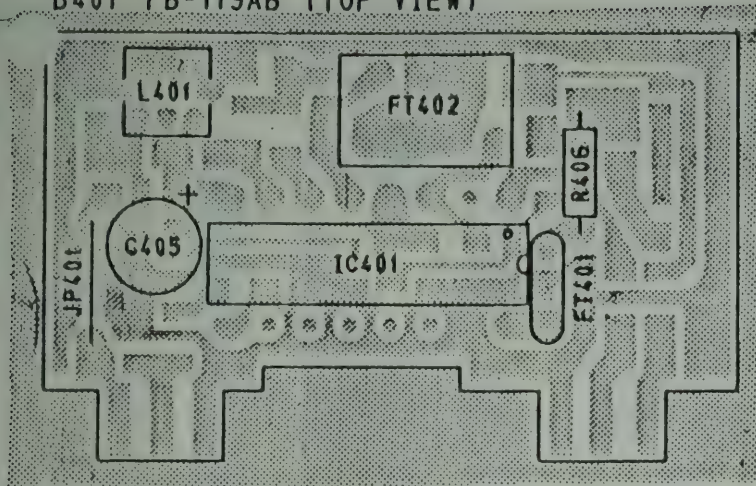


NOTES:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. 1K=KILO OHM. M=MEG OHM
2. RESISTOR WATTAGES ARE 1/8W UNLESS OTHERWISE NOTED.
3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. 1P=MICRO-MICRO FARAD

TITLE	NB PCB
PARTS ASSEMBLY	
E24-7567	REV. MARK

B401 PB-119AB (TOP VIEW)



L401	LF072

R406	470 1/6W

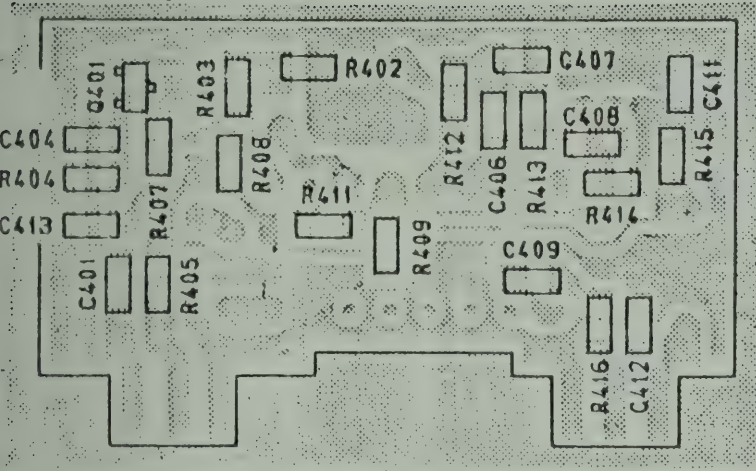
IC401	IR3N06

C405	10V47 C-095

FT401	FL048
FT402	FL066

JP401	7.5

(BOTTOM VIEW)



R402	12K
R403	22K
R404	1K
R405	1K
R407	1K
R408	100
R409	10
R411	2.2K
R412	2.2K
R413	47K
R414	68K
R415	100
R416	10K

C401	0.01/Y
C404	0.0047/X
C406	0.047/C
C407	0.047/C
C408	18P/RH
C409	0.047/C
C411	0.047/C
C412	0.047/C
C413	0.047/C

Q401	2SC2814F5

- NOTES:
1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. 1K-KILO OHM. M-MEG OHM)
 2. RESISTOR WATTAGES ARE 1/8W UNLESS OTHERWISE NOTED.
 3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. 1P-MICRO-MICRO FARAD

TITLE FM PCB	
PARTS ASSEMBLY	
E 24-7568	REV. MARK



B50/ PB112AA/LCD

IC501...IR2429

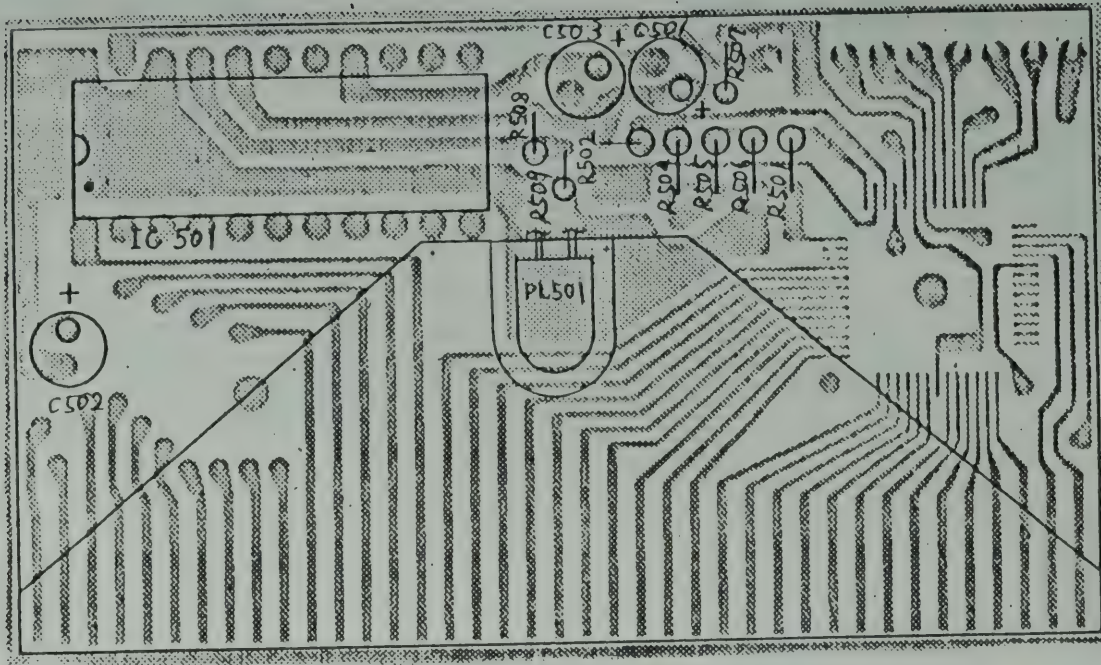
R501...220k 1/6W
R502...10k

R504...10k
R505...10k
R506...22k
R507...5k
R508...47k
R509...10k

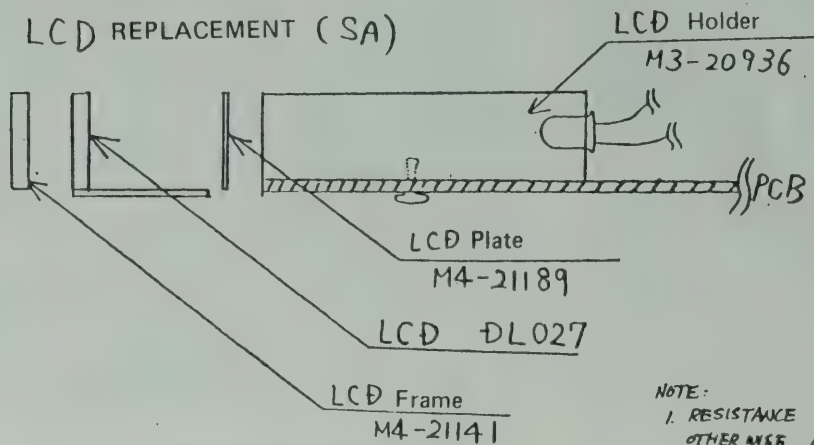
C501...10V 4.7
C502...50V 3.3
C503...6.3V 100

C503
C-156

PL501...PL102



LCD REPLACEMENT (SA)

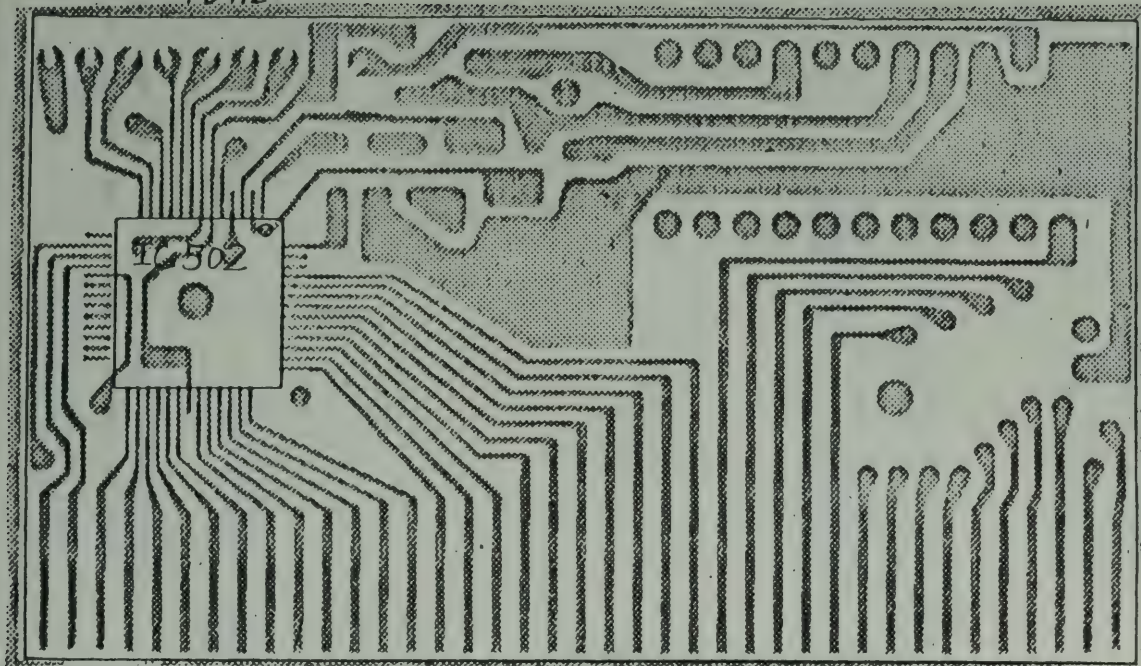


NOTE:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KILLOHM, M=MEGA OHM)
2. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P=PICTO-MICRO FARAD)

TITLE LCD PCB (TOP)	
PARTS ASSEMBLY	
DRAWING NO.	REV. MARK
E24-7569	

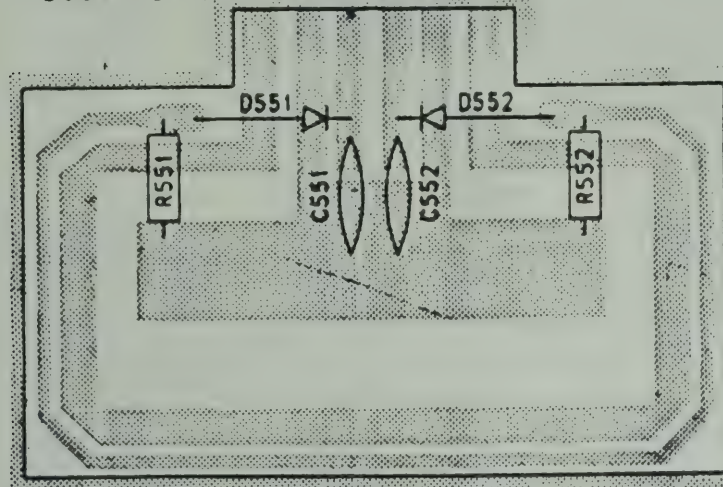
B501 PB112AA



IC502 LH5008 TP

TITLE LCD PCB (BOTTOM)	
PARTS ASSEMBLY	
DRAWING NO.	REV. MARK
E24-7570	

B551 PB118AA (TOP VIEW)



R551	220
R552	150
C551	0.01
C552	0.01
D551	1N60P
D552	1N60P

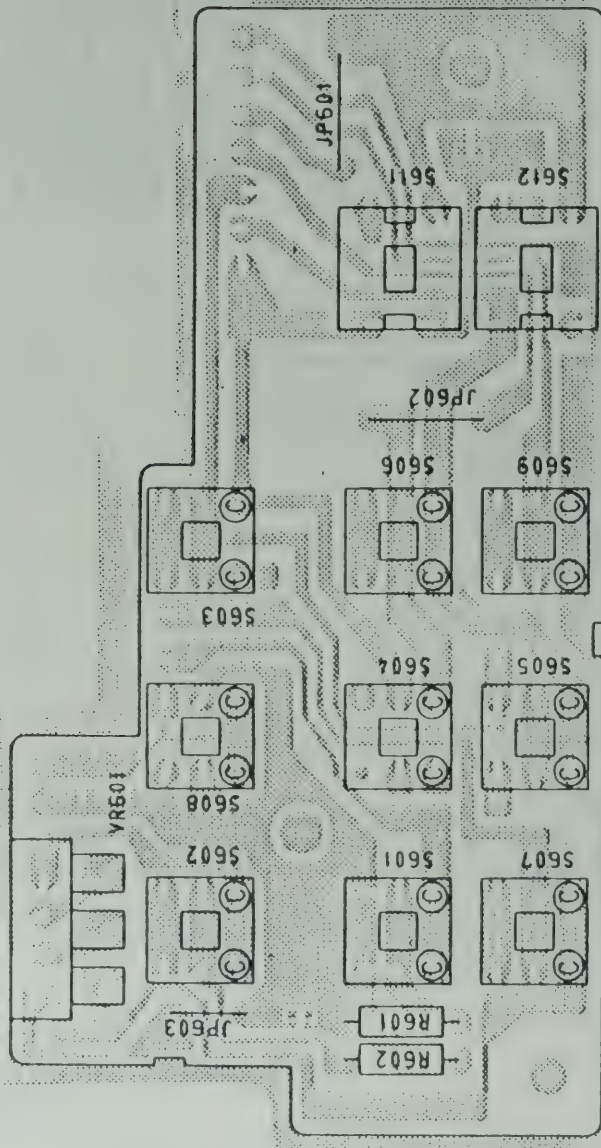
NOTES:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KILO OHM, M=MEG OHM)
2. RESISTOR WATTAGES ARE 1/6W UNLESS OTHERWISE NOTED.
3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P=MICRO-MICRO FARAD)
4. ALL CAPACITORS TEMPERATURE CHARACTERISTICS ARE YF UNLESS OTHERWISE NOTED.

TITLE SWR PCB	
PARTS ASSEMBLY TOP VIEW	
DRAWING NO.	REV. MARK
E24-7571	

B601 PB-1138A

UT550B/PB1138A::52:1



S601	SW549
S602	SW549
S603	SW629
S604	SW629
S605	SW629
S606	SW629
S607	SW549
S608	SW549
S609	SW549
S611	SW560
S612	SW560

R601	47 1/6W
R602	150 1/6W

VR601	RV672 5KB

JP601	7.5mm
JP602	7.5mm
JP603	5.0mm

NOTE:

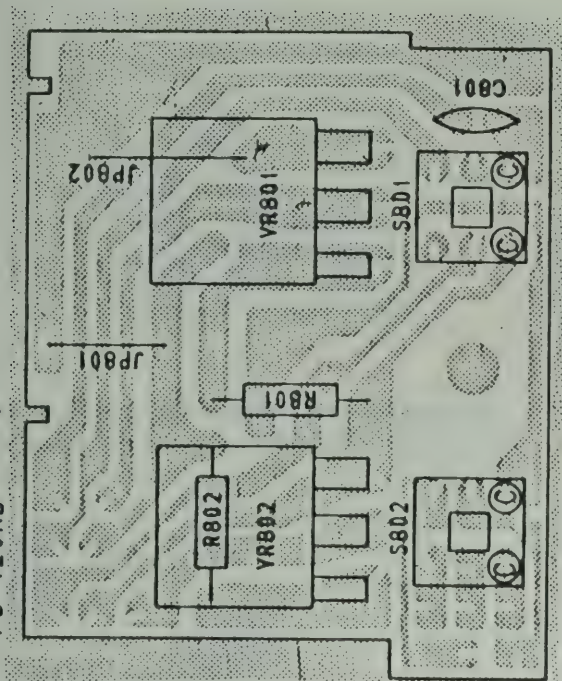
1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KILLO OHM. M=MEGA OHM)

TITLE	SW PCB
PARTS ASSEMBLY TOP VIEW	
REV.	
MARK	
E24-7572	

B 801

PB-120AB

UT550B/PB120AB::52:1



S801	SV549
S802	SV549

R801	2.2K
R802	1K

JP801	7.5mm
JP802	10mm

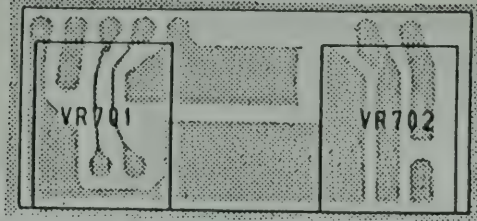
VR801	RV652 1KB
VR802	RV679 10KB

C801	0.022(SR)

- NOTES:
1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K-KILO OHM, M-MEG OHM)
 2. RESISTOR WATTAGES ARE 1/6W UNLESS OTHERWISE NOTED.
 3. CAPACITANCE VALUES ARE INDICATED IN MICRO FARADS UNLESS OTHERWISE NOTED. (P-MICRO-MICRO FARAD)

TITLE	RIT PCB
PARTS ASSEMBLY TOP VIEW	
REV.	
MARK	
E24-7573	

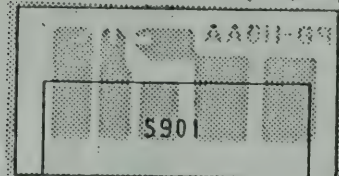
B701 PB-114AB (VoL pcb)



UT550/MIC.SW.CH.VOL::52:1

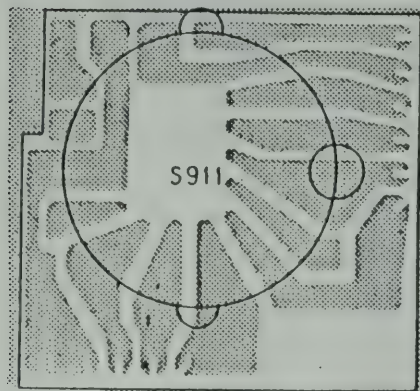
VR701	RV677 50KA
VR702	RV678 50KB

B901 PB-115AA (CH SW pcb)



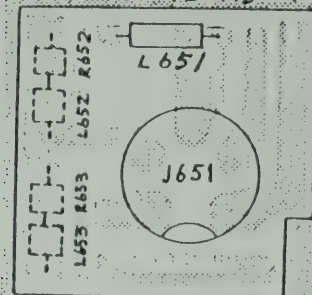
S901	SR404

B911 PB-116AA (MOBE SW pcb)



S911	SR405

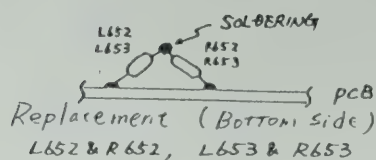
B651 PB-126AB (Mic JACK pcb)



J651	JK191
L651	LZ051 470μH
L652	LZ051 470μH
L653	LZ051 470μH
R652	1.5K 1/8 W
R653	1.5K 1/8 W

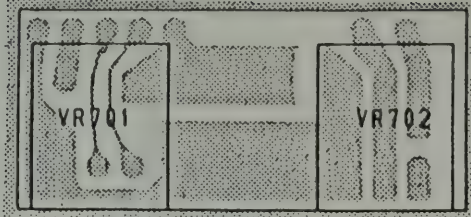
NOTE:

1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KIL OHM. M=MEG OHM)



TITLE	PARTS ASSEMBLY
	TOP VIEW
E24-7574	REV. MARK

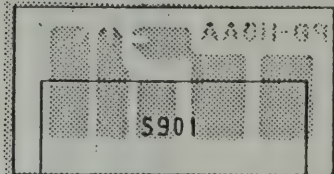
B701 PB-114AB (Vol PCB)



UT550/MIC.SW.CH.VOL::S2:1

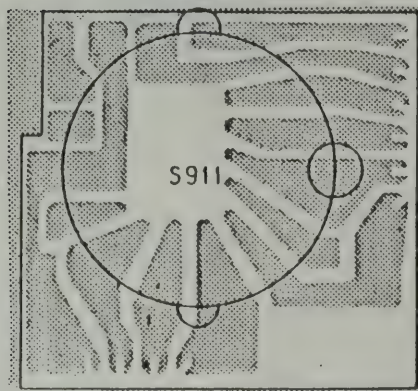
VR701	RV677 50KA
VR702	RV678 50KB

B901 PB-115AA (CH SW PCB)



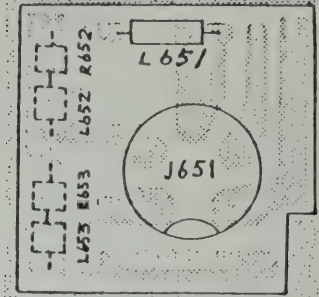
S901	SR404

B911 PB-116AA (MODE SW PCB)



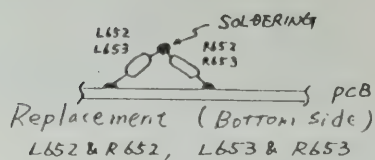
S911	SR405

B651 PB-126AB (MIC JACK PCB)

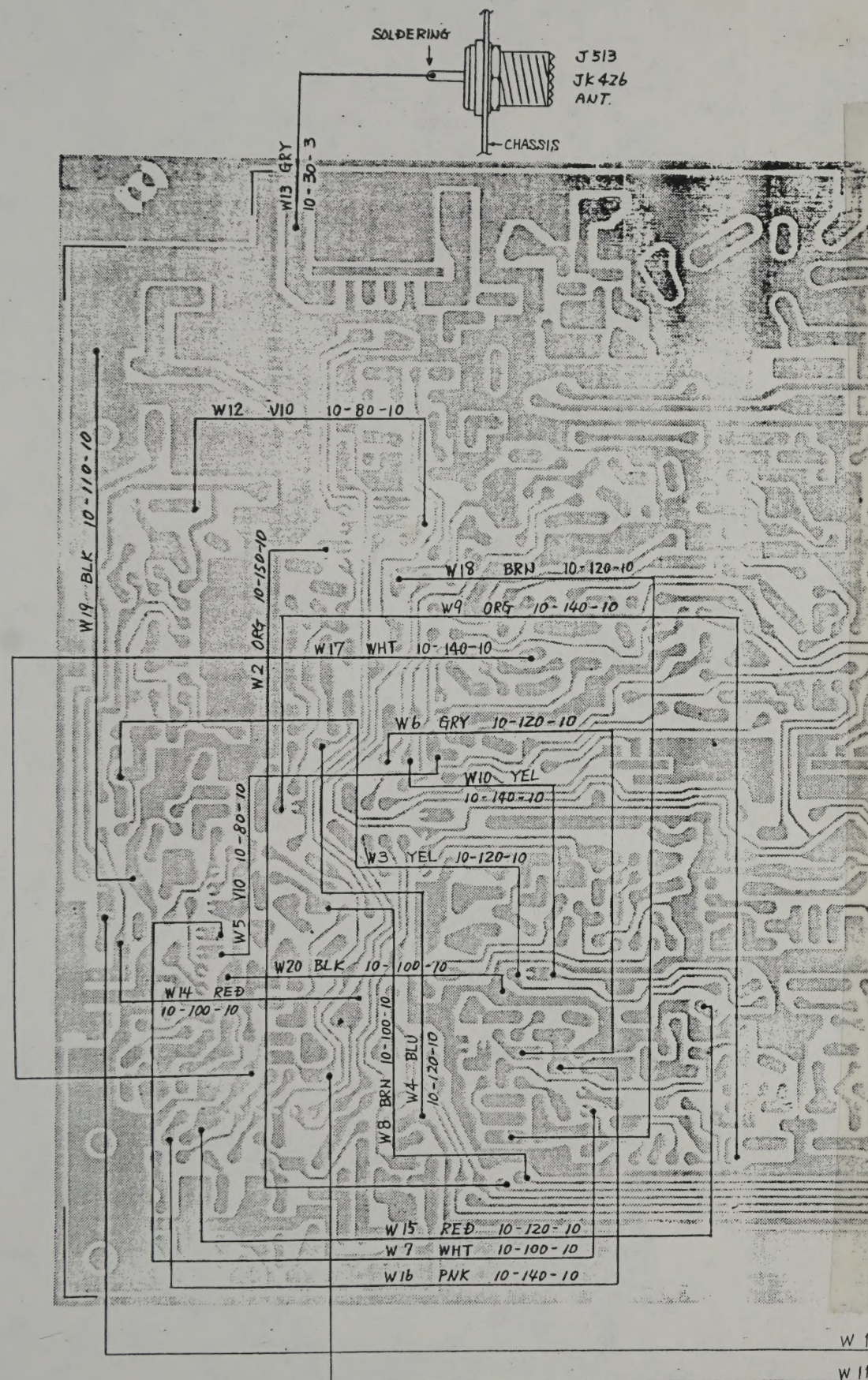


J651	JK191
L651	LZ051 470μH
L652	LZ051 470μH
L653	LZ051 470μH
R652	1.5K 1/6 W
R653	1.5K 1/6 W

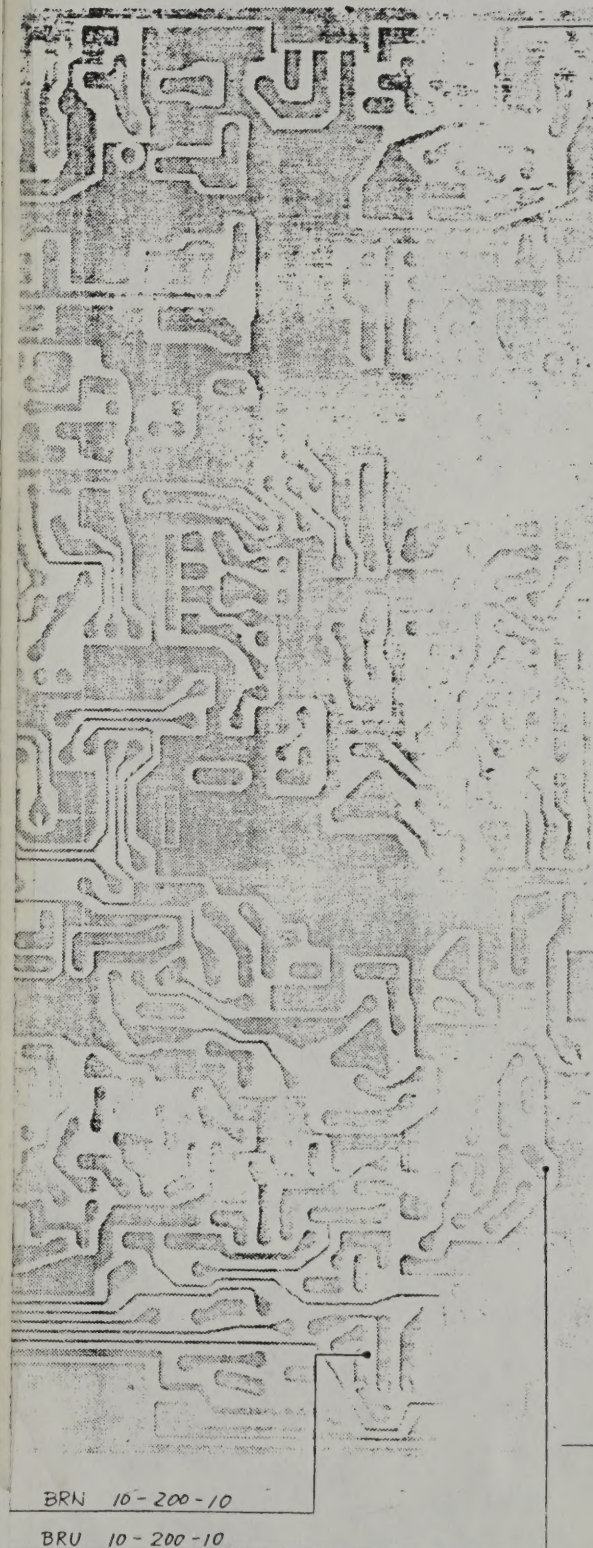
NOTE:
1. RESISTANCE VALUES ARE SHOWN IN OHMS UNLESS OTHERWISE NOTED. (K=KIL OHM. M=MEG OHM)

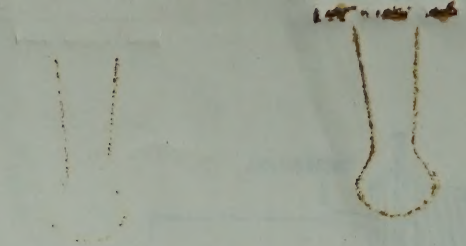


TITLE	PARTS ASSEMBLY
	TOP VIEW
E24-7574	REV. MARK



B001 PB-111 (TOP VIEW)





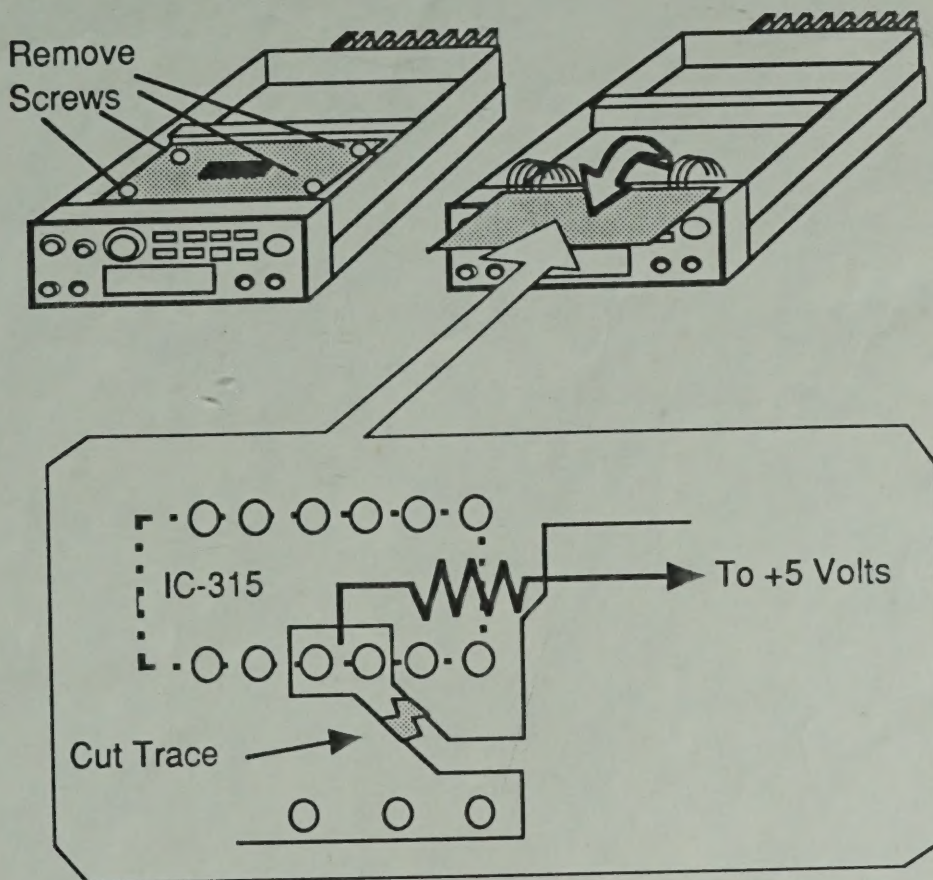
100-100000
100-100000

UNIDEN HR-2600

EXPANDED RF

You will need to replace the microprocessor. Replacement part # is

1. Remove Power and Antenna.
2. Remove screws and open the case.
3. Locate the Synthesizer board.
4. Pins 34 & 35 are grounded together on the underside of the synthesizer board. Cut the traces that connect these two pins to ground.
5. Solder one side of a 10K resistor to the connecting point of pins 34 & 35.
6. Connect the other leg of the 10 K resistor to + 5 volts. (where R181 & 187 are connected together).
7. Reassemble radio



COVERAGE : 26.0000 to 29.9999 MHz



Caution

These modifications have not been tested. The Author, Publisher and all other parties takes NO responsibility or liability for any damage or violation resulting from these modifications. Performing any modification may be a Violation of FCC Rules and will void the warranty of the radio. Use of any modified radio may be a violation of FCC rules. If you have any doubts, DO NOT PERFORM THIS MODIFICATION.
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EXPANDED R

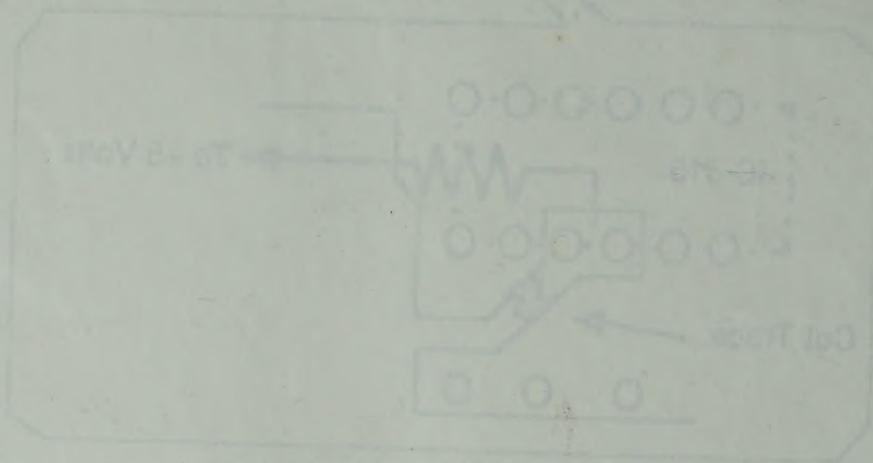
You will need to replace the microprocessor.
Part # is

UNIDEN

UC-1251

4045581

961 JAPAN



COVERAGE: 11,000 to 22,000 MHz

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